Maritime Sociology and the Blue Economy: Developing a Blue Sociology Framework

A Research Synthesis and Conceptual Framework Proposal

Executive Summary

This working paper proposes Blue Sociology as a complementary renewal and extension of maritime sociology toward an integrated sociology of sea societies in the context of accelerating Blue Economy development. Drawing on comprehensive literature synthesis across maritime sociology traditions, oceanic turn scholarship, and Blue Economy policy analysis, we argue that contemporary ocean governance faces critical social sustainability challenges that existing disciplinary frameworks inadequately address.

The Blue Sociology framework responds to three converging imperatives: (1) the rapid expansion of Blue Economy initiatives globally and within the EU, which intensify competing claims on marine spaces and resources; (2) persistent conceptual and methodological gaps in Blue Economy research and policy that marginalize social dimensions, justice concerns, and power asymmetries; and (3) theoretical innovations from the oceanic turn that reframe human-ocean relations through more-than-human, hydrosocial, and materialist ontologies.

We propose a tripartite analytical framework distinguishing:

- Marine domain: natural oceanographic and ecological processes
- Maritime domain: human activities, labor regimes, and institutional arrangements at sea
- Oceanic domain: planetary-scale socioecological systems and governance challenges

This framework integrates maritime sociology's empirical strengths in labor studies, organizational analysis, and territorial transformation with oceanic turn insights on multispecies entanglements, infrastructural politics, and hydrosocial relations. Through sectoral analysis of fisheries, ports, shipping, tourism, and offshore energy, we demonstrate how Blue Sociology reveals labor regimes, stratification patterns, institutional power dynamics, and legitimacy deficits that conventional Blue Economy frameworks obscure.

Analysis of EU Ocean Pact documents reveals fundamental tensions between economic growth imperatives and social sustainability goals, characterized by: superficial integration of social dimensions, technocratic governance rationalities that marginalize democratic participation, epistemic narrowness privileging natural science over social science and traditional knowledge, and unresolved contradictions between blue growth discourse and distributive justice requirements.

Blue Sociology addresses these gaps by foregrounding: multi-scalar socioecological integration, labor-centered social analysis across sectors, inclusion of more-than-human and infrastructural agencies, and democratic place-based governance mechanisms. The framework offers both analytical tools for critical research and practical guidance for equitable Blue Economy transitions.

Keywords: Blue Sociology, maritime sociology, Blue Economy, ocean governance, blue justice, labor regimes, oceanic turn, social sustainability, marine spatial planning, EU Ocean Pact

---

Table of Contents

1. Introduction: The Social Sustainability Challenge in Blue Economy Development
2. Theoretical Foundations: From Maritime Sociology to Blue Sociology
3. The Oceanic Turn: Contemporary Theoretical Innovations
4. EU Blue Economy Policy Framework: Tensions and Gaps
5. Sectoral Analysis: Social Dimensions Across Blue Economy Sectors
6. Research Gaps and Social Science Solutions
7. The Blue Sociology Framework: Analytical Architecture
8. Methodological Implications and Research Agenda
9. Policy Implications: Toward Just Blue Transitions
10. Conclusion: Blue Sociology as Complementary Renewal

---

1. Introduction: The Social Sustainability Challenge in Blue Economy Development {#1-introduction}

1.1 The Blue Economy Imperative and Its Discontents

The global ocean economy is undergoing rapid transformation. The European Union's Blue Economy Report documents nearly 5 million direct jobs and over €250 billion in annual gross value added from marine sectors [1]. The EU Ocean Pact positions oceans as critical for "life on Earth, prosperity, competitiveness, security, and a sustainable future," while acknowledging severe degradation from climate change, pollution, and overexploitation [2]. This blue growth agenda—expanding economic activities across fisheries, aquaculture, shipping, offshore energy, marine biotechnology, and coastal tourism—promises sustainable prosperity through technological innovation and integrated governance.

Yet mounting evidence reveals fundamental tensions between blue growth imperatives and social sustainability goals [3], [4], [5]. Small-scale fisheries face marginalization and dispossession as marine spaces are allocated to industrial uses [6], [7]. Coastal communities experience conflicts over access, participation deficits in planning processes, and unequal distribution of costs and benefits [8], [9]. Maritime labor remains characterized by precarity, weak enforcement of rights, and racialized hierarchies [10], [11]. Indigenous and local knowledge systems are systematically excluded from ocean governance [12], [13]. Gender dimensions of maritime work and coastal livelihoods receive superficial acknowledgment without substantive integration [14].

These social sustainability challenges are not incidental implementation problems but reflect deeper conceptual and methodological gaps in how Blue Economy is framed, researched, and governed. Dominant policy discourse privileges economic competitiveness and technocratic coordination while marginalizing questions of justice, power, and democratic participation [15], [16]. Natural science and technological innovation are positioned as primary knowledge sources, with limited integration of social science, humanities, or experiential knowledge from coastal communities [17]. Governance mechanisms favor sectoral representation and expert authority over participatory deliberation and community empowerment [18].

1.2 The Need for Blue Sociology

This working paper argues that addressing these challenges requires a renewed and extended sociological engagement with sea societies—what we term Blue Sociology. This framework builds on maritime sociology's long empirical and theoretical traditions while incorporating insights from the oceanic turn, environmental justice scholarship, and critical governance studies. Blue Sociology is not a replacement for maritime sociology but a complementary extension that responds to three specific imperatives:

First, the scale and scope of contemporary Blue Economy development demands integrated analysis across sectors, scales, and domains. Maritime sociology has traditionally focused on specific occupational communities (seafarers, fishers, port workers) and industries (shipping, fishing). While this sectoral depth remains essential, Blue Economy governance increasingly involves complex interactions across multiple uses of marine space, requiring analytical frameworks that can address cross-sectoral dynamics, cumulative impacts, and systemic transformations [19], [20].

Second, theoretical innovations from the oceanic turn offer crucial insights for understanding contemporary human-ocean relations. Scholarship on more-than-human geographies, hydrosocial relations, and materialist ontologies reframes oceans as relational, political, and lived spaces rather than inert resources [21], [22], [23]. These perspectives enable analysis of multispecies entanglements, infrastructural assemblages, and nonhuman agencies that shape social outcomes—dimensions largely absent from conventional Blue Economy frameworks but essential for socioecological sustainability [24], [25].

Third, the blue justice movement and critical scholarship on ocean governance have identified persistent equity deficits, power asymmetries, and democratic deficits in marine policy [26], [27], [28]. Addressing these requires analytical tools that can trace how political economy, institutional arrangements, and governance rationalities produce and reproduce inequalities—tools that maritime sociology and critical social science are well-positioned to provide [29], [30].

1.3 Research Approach and Evidence Base

This synthesis draws on comprehensive literature review across six thematic domains:

1. Sustainable Blue Economy research gaps (157 unique papers): Identifying conceptual and methodological deficits in Blue Economy research and policy

2. Maritimization theory and maritime sociology foundations (58 unique papers): Tracing theoretical traditions from Janiszewski through contemporary maritime sociology

3. Maritime work and organizations (60 unique papers): Analyzing labor regimes, organizational forms, and working conditions across maritime sectors

4. Blue sociology and oceanic turn perspectives (169 unique papers): Mapping contemporary theoretical innovations including more-than-human, hydrosocial, and blue urbanism approaches

5. EU Blue Economy sectoral analysis (172 unique papers): Examining fisheries, ports, shipping, tourism, and offshore energy with focus on social sustainability

6. Blue Economy labor and justice (177 unique papers): Investigating labor stratification, institutional power, legitimacy, and social justice dynamics

Additionally, we analyzed EU Ocean Pact policy documents to identify governance frameworks, policy priorities, and critical gaps in social dimension integration.

This evidence base enables systematic assessment of: (a) what maritime sociology and related social sciences already know about sea societies and maritime industries; (b) what conceptual and methodological innovations the oceanic turn offers; (c) what social sustainability challenges Blue Economy development presents; and (d) how Blue Sociology can address these challenges through integrated analytical frameworks.

1.4 Structure of This Paper

The paper proceeds as follows. Section 2 establishes theoretical foundations by synthesizing maritime sociology traditions and their contributions to understanding labor, organization, and territorial change. Section 3 maps the oceanic turn and its theoretical innovations. Section 4 critically analyzes EU Blue Economy policy frameworks, identifying tensions between growth imperatives and social sustainability goals. Section 5 presents sectoral analysis demonstrating how social dimensions manifest differently across fisheries, ports, shipping, tourism, and offshore energy. Section 6 systematically identifies research gaps and social science solutions. Section 7 proposes the Blue Sociology analytical framework, including the

tripartite Marine-Maritime-Oceanic architecture. Section 8 outlines methodological implications and research agenda. Section 9 discusses policy implications for just blue transitions. Section 10 concludes by positioning Blue Sociology as complementary renewal extending maritime sociology toward integrated sociology of sea societies.

---

2. Theoretical Foundations: From Maritime Sociology to Blue Sociology {#2-theoretical-foundations}

2.1 Maritime Sociology: Empirical Domains and Analytical Traditions

Maritime sociology emerged as a distinct subfield through sustained empirical engagement with seafaring occupations, shipboard social organization, port communities, and maritime industries [31], [32]. While never achieving the institutional consolidation of rural sociology or urban sociology, maritime sociology developed robust analytical traditions and accumulated substantial empirical knowledge about sea-based work and coastal societies [33].

Occupational sociology of seafaring: Early maritime sociology focused on seafarers as a distinctive occupational community characterized by spatial mobility, temporal rhythms of voyage and shore leave, and social isolation from land-based society [34], [35]. Fricke's seminal study of British merchant ship crews analyzed social structure, role differentiation, and organizational dynamics aboard vessels [36]. This tradition established methods for studying work organization in mobile, confined, and hierarchical settings—insights applicable to contemporary analysis of cruise ships, offshore platforms, and research vessels [37].

Labor process and employment relations: Maritime sociology contributed detailed analysis of labor processes, skill formation, and employment relations in shipping and fishing industries [38], [39]. Research documented how globalization transformed maritime labor markets through flags of convenience, international crewing, and regulatory arbitrage [40], [41]. Studies revealed racialized and nationalized hierarchies in crew composition, with officers from developed countries and ratings from developing nations [42], [43]. This political economy approach foregrounds power relations, profit imperatives, and regulatory gaps—analytical moves essential for understanding contemporary Blue Economy labor regimes [44].

Port communities and territorial transformation: Maritime sociology examined how ports shape urban development, regional economies, and territorial governance [45]. Vigarie's concept of "demaritimization" described how European port cities became disconnected from their maritime functions through containerization and spatial reorganization [46]. Recent work on "maritimization" analyzes how port expansion and maritime industries reshape territorial authority, land use, and sociotechnical trajectories [47], [48]. This territorial lens enables analysis of how Blue Economy development transforms coastal regions and urban-sea interfaces [49].

Organizational sociology and safety: Studies of shipboard organization, management styles, and occupational safety revealed how profit pressures, regulatory gaps, and organizational cultures shape working conditions and accident risks [50], [51]. Research on "lives in peril" documented systematic safety deficits in global shipping driven by cost competition and weak enforcement [52]. This critical organizational sociology provides frameworks for analyzing how institutional arrangements and power relations produce vulnerabilities—applicable to emerging offshore industries and aquaculture operations [53].

Social integration and community: Maritime sociology investigated how seafaring affects family life, community integration, and social bonds [54], [55]. Studies documented challenges of maintaining relationships across long absences, gendered divisions of domestic labor, and community support systems in maritime families [56]. This attention to social reproduction and community resilience informs contemporary analysis of how Blue Economy transformations affect coastal livelihoods and social cohesion [57].

2.2 Janiszewski's Maritimization Theory

Ludwik Janiszewski's concept of maritimization provides a crucial theoretical foundation for Blue Sociology [58]. Maritimization describes the process through which societies, territories, and economies become increasingly oriented toward and shaped by maritime activities. This is not merely economic dependence on marine resources but a deeper sociotechnical and cultural transformation involving:

- Territorial restructuring: How port development, maritime infrastructure, and ocean industries reshape land use, urban form, and regional governance
- Institutional adaptation: How legal frameworks, administrative structures, and governance mechanisms evolve to accommodate maritime activities
- Cultural orientation: How maritime identities, knowledge systems, and symbolic meanings become embedded in social life
- Sociotechnical trajectories: How technological systems (shipping, fishing, offshore platforms) co-evolve with social practices and institutional arrangements

Maritimization theory enables analysis of Blue Economy development as a long-term sociotechnical trajectory rather than discrete policy interventions [59]. It directs attention to path dependencies, lock-ins, and transformative moments when alternative trajectories become possible [60]. Applied to contemporary Blue Economy, maritimization theory illuminates how expanding ocean industries reshape coastal territories, governance institutions, and social relations—often in ways that concentrate power and marginalize traditional users [61].

2.3 Political Economy of Maritime Industries

Maritime sociology's political economy tradition analyzes how capitalist dynamics, state regulation, and global power relations shape maritime industries [62], [63]. Key insights include:

Flags of convenience and regulatory arbitrage: Shipping companies register vessels in states with minimal labor, safety, and environmental standards to reduce costs and avoid regulation [64]. This "race to the bottom" produces precarious employment, safety deficits, and environmental degradation while concentrating profits among shipowners and charterers [65]. The flag of convenience system exemplifies how global capitalism exploits regulatory fragmentation to externalize social and environmental costs [66].

Labor market segmentation: Maritime labor markets are highly segmented by nationality, race, and skill level, with systematic wage differentials and unequal access to training and advancement [67], [68]. Filipino, Indonesian, and Eastern European seafarers predominate in lower-wage ratings positions, while officers come disproportionately from developed countries [69]. This racialized division of labor reflects and reproduces global inequalities [70].

Profit imperatives and safety trade-offs: Research documents how profit maximization systematically undermines safety through cost-cutting on maintenance, crew size, training, and equipment [71]. Accidents and occupational health problems are not random but patterned by ownership structures, regulatory regimes, and competitive pressures [72]. This political economy lens reveals how market dynamics produce predictable social harms [73].

Weak collective organization: Seafarers face structural barriers to collective organization due to spatial dispersion, multinational crews, and employer strategies to fragment solidarity [74], [75]. Union density and collective bargaining power have declined with globalization, leaving workers vulnerable to exploitation [76]. Understanding these power asymmetries is essential for analyzing labor conditions across Blue Economy sectors [77].

2.4 Socioecological and Environmental Sociology Links

Recent maritime sociology has begun integrating environmental and socioecological perspectives, recognizing that ocean industries are embedded in and dependent upon marine ecosystems [78], [79]. Longo and colleagues' call for "marine sociology" emphasizes analyzing human-ocean relations through environmental sociology frameworks including treadmill of production, metabolic rift, and ecological modernization [80]. This socioecological turn positions maritime sociology to address:

- Social metabolism: Material and energy flows between society and marine environments, including extraction, pollution, and waste
- Ecological distribution conflicts: Struggles over access to marine resources and exposure to environmental harms
- Sustainability transitions: Pathways toward (or barriers to) socioecologically sustainable ocean use
- Climate change impacts: How ocean warming, acidification, and sea level rise affect maritime communities and industries

Integrating socioecological analysis with maritime sociology's labor and institutional focus enables Blue Sociology to address the coupled social-ecological dynamics that characterize Blue Economy development [81].

2.5 Contributions and Limitations

Maritime sociology provides Blue Sociology with:

- Empirical depth: Decades of research on maritime labor, organizations, and communities
- Analytical tools: Methods for studying work processes, organizational dynamics, and power relations
- Critical perspective: Political economy frameworks that foreground inequality, exploitation, and resistance
- Territorial lens: Concepts like maritimization for analyzing spatial transformation
- Socioecological foundations: Emerging integration of environmental sociology

However, maritime sociology also has limitations that Blue Sociology must address:

- Sectoral fragmentation: Traditional focus on specific industries limits cross-sectoral analysis
- Anthropocentrism: Limited engagement with more-than-human perspectives and nonhuman agencies
- Scale gaps: Insufficient attention to planetary-scale ocean processes and governance
- Governance deficits: Underdeveloped analysis of participatory governance and democratic legitimacy
- Justice frameworks: Limited integration of environmental justice and blue justice scholarship

Blue Sociology builds on maritime sociology's strengths while addressing these limitations through integration with oceanic turn perspectives, governance studies, and justice frameworks.

---

3. The Oceanic Turn: Contemporary Theoretical Innovations {#3-the-oceanic-turn}

3.1 Defining the Oceanic Turn

The oceanic turn in social sciences and humanities represents a fundamental reorientation in how scholars conceptualize and study oceans and human-ocean relations [82], [83]. Moving beyond treating oceans as inert resources, transportation corridors, or territorial spaces to be governed, oceanic turn scholarship frames seas as:

- Relational spaces: Constituted through ongoing interactions among humans, nonhumans, materials, and forces
- Political arenas: Sites of contestation, power struggles, and competing claims

- Lived environments: Experienced, inhabited, and meaningful places for diverse communities
- Multispecies assemblages: Entanglements of human and nonhuman life, technologies, and ecologies
- Planetary systems: Interconnected with atmospheric, terrestrial, and cryospheric processes at global scales

This reframing draws on theoretical innovations from human geography, science and technology studies, environmental humanities, and political ecology [84], [85]. The oceanic turn challenges terrestrial bias in social theory and develops concepts, methods, and sensibilities attuned to aquatic specificities [86].

3.2 More-Than-Human Geographies and Multispecies Entanglements

More-than-human geographies extend analysis beyond human actors to include nonhuman animals, organisms, materials, and forces as active participants in social-ecological processes [87], [88]. Applied to oceans, this perspective reveals:

Multispecies fisheries: Fishing is not simply human extraction of resources but complex entanglements among fishers, fish, boats, gear, weather, markets, and regulations [89]. Fish behavior, migration patterns, and population dynamics shape fishing practices and livelihoods as much as human decisions shape fish populations [90]. Understanding fisheries requires attending to these multispecies relations rather than treating fish as passive objects [91].

Marine protected areas as assemblages: MPAs are not just regulatory boundaries but assemblages of marine life, human users, enforcement technologies, scientific knowledge, and governance institutions [92]. Conservation outcomes emerge from interactions among these heterogeneous elements, not from policy design alone [93]. More-than-human analysis reveals how nonhuman agencies (coral growth, fish movement, algal blooms) enable or constrain conservation goals [94].

Offshore energy infrastructures: Wind turbines, oil platforms, and tidal generators create novel marine habitats, alter hydrodynamics, and attract or repel marine life [95]. These infrastructures are not simply human impositions on nature but become integrated into marine ecosystems, producing hybrid socionatural assemblages [96]. Analyzing offshore energy requires attending to these material-ecological entanglements [97].

Blue Economy as more-than-human: Bear's analysis of "assembling ocean life" in the Blue Economy demonstrates how economic development depends on enrolling nonhuman actors—fish, seaweed, microbes, minerals—into commodity chains and value networks [98]. Blue Economy is thus fundamentally a more-than-human project, though typically framed in anthropocentric terms [99].

3.3 Hydrosocial Relations and Aquatic Ontologies

Hydrosocial approaches reconceptualize water (including seawater) as simultaneously material and social, natural and political [100], [101]. Water is not a pre-existing resource that societies manage but is produced through sociotechnical practices, infrastructures, and power relations [102]. Applied to oceans, hydrosocial thinking emphasizes:

Oceanic flows and circulations: Ocean currents, tides, and waves are not merely physical phenomena but shape and are shaped by human activities—navigation routes, fishing grounds, pollution dispersal, climate patterns [103]. Hydrosocial analysis traces how these flows connect distant places and actors in relations of interdependence and vulnerability [104].

Maritime infrastructures: Ports, shipping lanes, submarine cables, pipelines, and offshore platforms are hydrosocial infrastructures that channel and control oceanic flows while being shaped by oceanic forces [105]. These infrastructures embody power relations, enabling some actors to harness oceanic flows while constraining others [106].

Territorial waters and jurisdictions: Maritime boundaries and exclusive economic zones are hydrosocial productions—attempts to impose terrestrial logics of bounded territory onto fluid, dynamic ocean spaces [107]. The tensions and contradictions in ocean governance partly reflect this mismatch between territorial governance and aquatic ontologies [108].

Wet ontologies: Some scholars propose "wet ontologies" that take fluidity, permeability, and flow as fundamental rather than solidity and boundedness [109]. This challenges terrestrial assumptions embedded in social theory and governance frameworks, opening space for alternative conceptualizations more attuned to oceanic specificities [110].

3.4 Materialist and Infrastructural Approaches

New materialist perspectives emphasize the agency of materials, objects, and infrastructures in shaping social life [111]. Applied to oceans, materialist approaches analyze:

Oceanic materialities: Salt, pressure, temperature, viscosity, and other material properties of seawater shape what is possible and impossible in ocean spaces [112]. These materialities constrain and enable human activities, from swimming to shipping to offshore construction [113]. Materialist analysis attends to how oceanic matter actively participates in social processes [114].

Maritime technologies: Ships, fishing gear, diving equipment, sensors, and other technologies are not neutral tools but active mediators that shape human-ocean relations [115]. Technological systems embody particular ways of knowing and using oceans, privileging some practices while marginalizing others [116]. Critical analysis of maritime technologies reveals power relations and path dependencies [117].

Infrastructural politics: Maritime infrastructures—ports, cables, pipelines, platforms—are political in multiple senses: they embody decisions about resource allocation, they enable some activities while constraining others, and they become sites of contestation [118]. Infrastructural analysis reveals how material arrangements shape social possibilities and power relations [119].

Plastic oceans: The accumulation of plastic pollution exemplifies how human materials become integrated into marine ecosystems, creating novel socionatural assemblages with unpredictable consequences [120]. Plastic pollution is not just an

environmental problem but a materialist challenge requiring analysis of production systems, consumption patterns, and waste infrastructures [121].

3.5 Blue Urbanism and Coastal Assemblages

Blue urbanism examines how cities relate to adjacent oceans, seas, and coasts, challenging the terrestrial bias in urban studies [122], [123]. Key themes include:

Urban-ocean interfaces: Waterfronts, beaches, ports, and coastal infrastructure are hybrid zones where urban and oceanic processes intersect [124]. Blue urbanism analyzes how these interfaces are designed, governed, and experienced, and how they mediate urban-ocean relations [125].

Marine protected areas in cities: Urban MPAs represent attempts to integrate conservation into densely populated coastal areas [126]. Research shows these are rare but potentially valuable for both biodiversity and community well-being when designed through participatory processes [127]. Urban MPAs exemplify blue democracy—linking coastal communities to marine governance [128].

Coastal gentrification and displacement: Waterfront redevelopment often produces gentrification, displacing working-class and maritime communities while creating exclusive spaces for affluent residents and tourists [129]. Blue urbanism critically analyzes these processes of spatial transformation and social exclusion [130].

Amphibious urbanism: Some scholars propose "amphibious" approaches to coastal cities that embrace rather than resist oceanic dynamics, designing for flooding, tides, and sea level rise [131]. This requires rethinking urban planning assumptions and developing governance capacities for living with water [132].

3.6 Political Ecology of Oceans

Political ecology analyzes how power relations, political economy, and cultural politics shape environmental conditions and resource access [133]. Marine political ecology applies these frameworks to oceans:

Ocean grabbing: Analogous to land grabbing, ocean grabbing describes processes through which powerful actors (states, corporations, conservation organizations) appropriate marine spaces and resources, displacing or marginalizing existing users [134], [135]. Blue Economy development often involves ocean grabbing dynamics as marine areas are allocated to industrial uses or conservation zones [136].

Ecological distribution conflicts: Struggles over who benefits from marine resources and who bears environmental burdens are fundamentally political [137]. Political ecology analyzes how these conflicts reflect and reproduce social inequalities along lines of class, race, gender, and geography [138].

Discourse and knowledge politics: How oceans are represented—as wilderness, resources, commons, property—shapes governance possibilities [139]. Political ecology examines whose knowledge counts in ocean governance and how dominant discourses (like blue growth) naturalize particular interests while marginalizing alternatives [140].

Scalar politics: Ocean governance involves contestation over appropriate scales of decision-making—local, national, regional, global [141]. Political ecology analyzes how scalar framings privilege certain actors and interests while excluding others [142].

3.7 Indigenous and Decolonial Ocean Scholarship

Indigenous ocean scholarship centers Indigenous peoples' relationships with oceans, challenging colonial appropriations and asserting Indigenous sovereignty and stewardship [143], [144]. Key contributions include:

Indigenous marine tenure: Many Indigenous peoples have sophisticated systems of marine tenure, resource management, and ecological knowledge developed over millennia [145]. These systems are often more sustainable and equitable than state-imposed regimes but have been systematically undermined by colonization [146].

Relational ontologies: Indigenous ocean philosophies often emphasize kinship, reciprocity, and responsibility rather than property and extraction [147]. These relational ontologies offer alternatives to dominant Western frameworks that treat oceans as resources to be exploited [148].

Decolonizing ocean governance: Decolonial scholars critique how international ocean law and governance institutions perpetuate colonial power relations and call for fundamental restructuring to recognize Indigenous rights and sovereignty [149], [150]. This includes implementing free, prior, and informed consent for marine developments affecting Indigenous territories [151].

Oceanic epistemologies: Indigenous knowledge systems include sophisticated understanding of ocean processes, marine life, and sustainable use practices [152]. Decolonial approaches demand these epistemologies be recognized as legitimate knowledge rather than dismissed as "traditional" or "local" in contrast to "scientific" knowledge [153].

3.8 Feminist Ocean Geographies

Feminist scholarship on oceans reveals gendered dimensions of maritime work, coastal livelihoods, and ocean governance [154], [155]. Key insights include:

Gendered divisions of labor: Maritime industries are highly gender-segregated, with men predominating in fishing, seafaring, and offshore work while women concentrate in shore-based processing, administration, and service roles [156]. These divisions reflect and reproduce gender inequalities in pay, status, and decision-making power [157].

Invisible labor: Women's contributions to maritime livelihoods—mending nets, processing catch, managing household economies during men's absences—are often unrecognized and uncompensated [158]. Feminist analysis makes this invisible labor visible and demands its recognition in policy [159].

Embodied experiences: Feminist scholarship attends to embodied, affective, and emotional dimensions of ocean relations often absent from masculinist accounts [160]. This includes experiences of risk, loss, care, and connection that shape maritime communities [161].

Intersectionality: Feminist ocean geographies analyze how gender intersects with race, class, nationality, and other axes of difference to produce varied experiences of maritime work and coastal life [162]. This intersectional lens reveals how power operates through multiple, overlapping systems of oppression [163].

3.9 Contributions to Blue Sociology

The oceanic turn provides Blue Sociology with:

- Ontological reorientation: Frameworks for understanding oceans as relational, political, and lived rather than inert resources
- Analytical expansion: Concepts for analyzing multispecies entanglements, material agencies, and infrastructural politics
- Methodological innovation: Approaches attuned to fluidity, flow, and aquatic specificities
- Justice frameworks: Tools for analyzing power, inequality, and marginalization in ocean contexts
- Decolonial critique: Challenges to colonial legacies and Western-centric ocean governance
- Feminist insights: Attention to gender, embodiment, and invisible labor

Integrating these oceanic turn innovations with maritime sociology's empirical depth and analytical traditions enables Blue Sociology to develop comprehensive frameworks for understanding and transforming contemporary sea societies.

---

4. EU Blue Economy Policy Framework: Tensions and Gaps {#4-eu-blue-economy-policy-framework}

4.1 The EU Ocean Pact: Priorities and Architecture

The European Ocean Pact establishes a comprehensive framework positioning the EU as a global leader in ocean governance [164]. The pact identifies oceans as critical for prosperity, competitiveness, security, and sustainability while acknowledging severe degradation from climate change, pollution, and overexploitation [165]. Core policy priorities include:

Protection and restoration: Safeguarding and restoring ocean health, marine biodiversity, and biological resources through addressing pollution and promoting sustainable practices [166], [167].

Blue Economy competitiveness: Fostering "a competitive and sustainable blue economy" across traditional sectors (fisheries, aquaculture) and emerging industries (marine renewables, biotechnology) [168].

Coastal community support: Enhancing resilience and economic autonomy of coastal and island communities, particularly in outermost regions, through sustainable tourism and new business models [169].

Ocean research and knowledge: Advancing research, innovation, skills development, and ocean observation capabilities [170].

Maritime security: Strengthening maritime security and defense capabilities, including protection of underwater infrastructure and freedom of navigation [171].

International governance: Upholding UNCLOS, ratifying the BBNJ agreement, combating IUU fishing, and advocating for global marine pollution treaties [172].

The institutional framework proposes a "unified reference framework" to improve coordination, an "Ocean Act" based on reviewing the Marine Spatial Planning Directive, a "high-level Ocean Council" with sectoral representatives, and a monitoring dashboard using existing data systems [173], [174].

4.2 Marginalization of Social Dimensions

Critical analysis reveals systematic marginalization of social dimensions across the Ocean Pact [175]:

Labor and working conditions: The pact frames labor primarily through economic contribution metrics (5 million jobs, €250 billion value added) rather than worker rights or decent work frameworks [176]. There is no mention of labor rights, occupational safety, precarious employment, ILO conventions, automation impacts, or migrant labor issues [177]. A single reference to "improving working conditions" in fishing provides no specific mechanisms or accountability measures [178].

Community participation: While the pact claims to "involve all relevant stakeholders," the list prioritizes "innovators" and "investors" over fishing communities or coastal populations [179]. There is no specification of participatory mechanisms, decision-making power, co-management arrangements, or free, prior, and informed consent principles [180]. The proposed Ocean Council comprises "sectoral representatives" without clarifying how civil society or small-scale users will be represented [181].

Distributive justice: Despite rhetoric about "social inclusion, justice, and solidarity," these terms remain undefined and unoperationalized [182]. The pact lacks frameworks for assessing distributional impacts, ensuring equitable benefit-sharing from marine resources, or addressing who bears costs versus who captures benefits [183]. There is no consideration of spatial justice—how marine spatial planning affects different users' access [184].

Traditional and Indigenous knowledge: This represents perhaps the most glaring gap [185]. There is no mention of traditional knowledge, indigenous knowledge, local ecological knowledge, traditional fishing practices, customary marine tenure, or ancestral connections to marine spaces [186]. The emphasis on "science-based decision-making" privileges Western scientific knowledge without acknowledging complementary knowledge systems [187].

Gender dimensions: One sentence acknowledges the need to strengthen women's position in ocean activities [188]. This tokenistic treatment provides no gender analysis, gender-disaggregated data, gender impact assessment frameworks, or consideration of how policies differentially affect women and men [189]. Women are framed through a deficit model rather than addressing structural barriers [190].

Power asymmetries: The pact fails to examine power relations shaping ocean governance [191]. There is no acknowledgment of corporate concentration, lobbying influence, market power, North-South dynamics, or financialization effects [192]. By treating oceans primarily as economic opportunity spaces, the pact naturalizes existing power structures [193].

4.3 Tensions Between Growth and Sustainability

The Ocean Pact embodies fundamental tensions between economic growth imperatives and social-environmental sustainability commitments [194]:

Blue growth paradigm: The pact aims to "boost competitiveness" while ensuring sustainability, reflecting ecological modernization assumptions that growth and environmental protection are compatible through innovation [195]. This raises critical questions: Can continuous growth in ocean activities be reconciled with ocean health restoration? Does growth imperative override sustainability when conflicts arise? Who defines "sustainable"? [196]

Spatial competition: The pact acknowledges "increasing pressure on limited marine areas" from multiple activities but assumes compatibility through better management rather than recognizing fundamental conflicts [197]. Expanding multiple industries simultaneously while protecting 30% of marine areas creates spatial and ecological contradictions [198]. Who loses access when areas are allocated to industrial uses or conservation? [199]

Fisheries modernization: Seeking to "increase competitiveness" while "maintaining stable supplies" and "applying ecosystem-based approach" creates tensions [200]. Fleet modernization often means larger vessels that displace small-scale fishers [201]. Competitiveness pressures favor industrial operations over artisanal fisheries [202]. Ecosystem-based management may restrict fishing in ways that differentially impact small-scale versus industrial operators [203].

Resource exploitation versus conservation: Highlighting oceans as sources of genetic resources and minerals while stressing environmental protection creates contradictions [204]. Marine genetic resources raise biopiracy concerns [205]. Mineral extraction has uncertain ecological impacts [206]. Who benefits from commercializing marine resources versus who bears costs? [207]

Decarbonization versus competitiveness: Emphasizing maritime sector decarbonization while maintaining competitiveness creates tensions [208]. Decarbonization requires substantial investment that may disadvantage smaller operators [209]. Green technology transition may concentrate market power among actors with capital access [210]. Carbon pricing impacts may be regressive [211].

4.4 Technocratic Governance and Democratic Deficits

The proposed governance mechanisms reveal technocratic rationalities that marginalize democratic participation [212]:

Technocratic bias: The architecture privileges technical expertise, sectoral representation, and administrative efficiency over democratic participation and deliberation [213]. High-level councils of sectoral representatives may exclude non-organized stakeholders [214]. Dashboard monitoring emphasizes quantifiable indicators, potentially missing qualitative social dimensions [215]. "Streamlining" may reduce opportunities for public input [216].

Accountability gaps: It is unclear how the Ocean Council will be accountable to affected communities [217]. There is no specification of transparency mechanisms or public access to decision-making [218]. Missing provisions for independent oversight or social impact evaluation [219].

Top-down orientation: The pact emphasizes EU-level coordination with limited attention to subsidiarity or local governance [220]. EU-level frameworks may be insensitive to regional variations [221]. Centralized governance can disempower local communities and traditional management systems [222].

Sectoral representation model: Based on the European Council for Agriculture and Food, this model suggests sectoral representation [223]. Sectoral approaches may perpetuate silos despite integration rhetoric [224]. Organized sectors have more representation capacity than diffuse interests [225]. Risk of regulatory capture by well-resourced sectoral actors [226].

Knowledge politics: Positioning natural science and technology as primary knowledge sources excludes other knowledge forms [227]. Science is not neutral; research agendas reflect funding and power relations [228]. Technical knowledge alone cannot resolve normative questions about values and justice [229].

4.5 Financial Governance and Investment Logic

The pact's emphasis on mobilizing public and private investment raises governance concerns [230]:

Investment-centric approach: Heavy emphasis on "significant public and private investment" and the "BlueInvest platform" for start-ups and SMEs [231]. Financialization of ocean governance may prioritize investor returns over social or environmental outcomes [232]. Public-private partnerships may transfer public resources to private profit [233].

Conditionality and autonomy: Public funding leverage for private investment may prioritize investor interests [234]. Conditionality attached to EU funds can constrain Member State and local autonomy [235]. Financial mechanisms may favor large-scale projects over small-scale initiatives [236].

Blue washing risks: Unclear how social and environmental standards will be enforced for funded activities [237]. Risk of projects marketed as sustainable without substantive standards [238].

4.6 Missing Paradigms and Alternative Framings

The Ocean Pact's blue growth framing marginalizes alternative paradigms [239]:

Rights-based approaches: Conspicuously absent are rights-based framings—rights of nature, human rights to healthy ocean environment, rights of coastal communities to livelihoods and culture, rights of future generations [240].

Commons governance: Despite rich scholarship on governing common-pool resources, the pact shows limited engagement with commons principles—community-based management, polycentric governance, traditional tenure systems, collective action beyond market and state [241].

Degrowth and post-growth: The pact doesn't engage with debates about degrowth (planned reduction of resource throughput), post-growth (economies designed for well-being rather than GDP growth), or sufficiency (meeting needs through less consumption) [242].

Transformative change: The pact proposes incremental improvements within existing structures rather than engaging with sustainability science emphasis on transformative change—fundamental shifts in values, power relations, and institutions [243].

4.7 Synthesis: A Framework Requiring Social Science Critique

The EU Ocean Pact represents an ambitious coordination effort but reveals significant limitations:

1. Comprehensive scope, narrow framing: Addresses multiple ocean issues but within blue growth paradigm privileging economic competitiveness and technocratic governance [244]

2. Social dimensions as afterthought: Labor, participation, justice, traditional knowledge, and gender receive superficial acknowledgment without substantive integration [245]

3. Unresolved tensions: Growth imperatives and sustainability goals remain in tension, papered over with win-win rhetoric [246]
4. Democratic deficits: Limited capacity for democratic participation, accountability, and addressing power asymmetries [247]

5. Epistemic narrowness: Heavy reliance on natural science with limited social science, humanities, or traditional knowledge integration [248]

These limitations are not incidental but reflect deeper paradigmatic commitments to blue growth, ecological modernization, and technocratic governance. Addressing them requires the kind of critical social science analysis that Blue Sociology provides—foregrounding political economy, power relations, justice frameworks, and alternative paradigms for human-ocean relations.

---

5. Sectoral Analysis: Social Dimensions Across Blue Economy Sectors {#5-sectoral-analysis}

5.1 Introduction: Sectoral Diversity and Common Patterns

Blue Economy encompasses diverse sectors—fisheries, aquaculture, shipping, ports, offshore energy, tourism, marine biotechnology—each with distinct social dynamics, labor regimes, and governance challenges [249]. Yet common patterns emerge: labor precarity and stratification, power asymmetries between industrial and small-scale actors, participation deficits in governance, distributional conflicts over access and benefits, and tensions between economic development and social sustainability [250].

This section applies maritime sociology and oceanic turn lenses to five major sectors, demonstrating how Blue Sociology reveals social dimensions that conventional Blue Economy frameworks obscure. The analysis draws on the comprehensive literature synthesis to identify sector-specific manifestations of labor regimes, institutional power, legitimacy dynamics, and justice concerns.

5.2 Fisheries: Access, Justice, and Livelihood Struggles

Social stratification and access regimes: Fisheries are characterized by profound inequalities between industrial and small-scale operations [251], [252]. Large-scale industrial fishing fleets, often vertically integrated with processing and distribution, dominate access to fish stocks and markets [253]. Small-scale and artisanal fishers face marginalization through multiple mechanisms: allocation of quotas favoring large operators, spatial exclusion from productive fishing grounds, market power of buyers and processors, and limited political influence in fisheries governance [254], [255].

Historical analysis reveals that equitable access depends on institutional design [256]. Community-based management systems, when genuinely empowered with authority and resources, can maintain equitable access and sustainable practices [257]. However, neoliberal fisheries reforms emphasizing individual transferable quotas and market-based allocation often concentrate fishing rights among capital-rich actors, displacing small-scale fishers [258], [259].

Labor conditions and precarity: Fishing remains one of the most dangerous occupations globally, with high rates of injury and mortality [260]. Labor conditions vary dramatically by scale and regulatory context. Industrial fishing vessels, particularly those operating under flags of convenience, often employ migrant workers under exploitative conditions with minimal safety protections [261]. Small-scale fishers face economic precarity from volatile catches, market fluctuations, and increasing costs [262].

Gender dimensions are significant but often invisible [263]. Women predominate in shore-based activities—processing, marketing, net mending—that are essential to fishing livelihoods but undervalued and poorly compensated [264]. Women's exclusion from decision-making in fisheries governance perpetuates gender inequalities [265].

Blue Economy pressures: Blue Economy development intensifies pressures on small-scale fisheries through multiple pathways [266], [267]. Marine spatial planning allocates areas to aquaculture, offshore energy, shipping lanes, and conservation zones, reducing space available for fishing [268]. Coastal tourism development transforms fishing harbors into marinas and waterfronts, displacing fishing communities through gentrification [269]. Industrial aquaculture competes for coastal space and may degrade water quality affecting wild fisheries [270].

Justice frameworks and resistance: Blue justice scholarship frames fisheries conflicts as struggles over rights, recognition, and redistribution [271], [272]. Small-scale fishers and their organizations increasingly mobilize around rights-based frameworks, demanding recognition of customary rights, equitable benefit-sharing, and meaningful participation in governance [273], [274]. The FAO Voluntary Guidelines for Securing Sustainable Small-Scale Fisheries provide normative framework, though implementation remains weak [275].

Research documents how fisher movements challenge blue economy industrialization, asserting alternative visions of ocean use centered on food sovereignty, cultural continuity, and ecological stewardship [276], [277]. These struggles exemplify broader tensions between blue growth and social justice imperatives [278].

5.3 Ports: Territorial Transformation and Labor Restructuring

Maritimization and territorial change: Ports are sites of intensive maritimization—territorial transformation driven by maritime industries [279]. Port expansion reshapes urban form, land use, and regional economies [280]. Containerization and automation have fundamentally restructured port operations, reducing labor requirements while increasing throughput [281].

European port cities experienced "demaritimization"—disconnection between cities and their maritime functions as ports relocated to peripheral sites and traditional maritime communities declined [282]. Contemporary port development often involves conflicts over land use, environmental impacts, and community displacement [283].

Labor regime transformation: Port labor has undergone dramatic restructuring over recent decades [284]. Traditional dock work, characterized by strong unions, relatively high wages, and craft skills, has been largely displaced by containerization and automation [285]. Contemporary port labor is increasingly casualized, with workers employed through temporary agencies rather than permanent positions [286].

Multi-skilling and flexibility demands have eroded traditional occupational identities and collective organization [287]. Port workers face intensified work pace, surveillance, and performance pressures [288]. Union density and bargaining power have declined, though pockets of strong organization persist [289].

Governance and legitimacy: Port governance involves complex interactions among port authorities, municipal governments, national regulators, shipping companies, and logistics firms [290]. Ports are often governed as "exceptional spaces" with special regulatory regimes that can override local planning and environmental protections [291].

Community participation in port governance is typically weak [292]. Port expansion decisions are made through technocratic processes privileging economic and logistical considerations over social and environmental concerns [293]. This produces legitimacy deficits and community opposition to port development [294].

Blue Economy implications: Ports are critical infrastructure for Blue Economy development, serving fisheries, offshore energy, cruise tourism, and marine industries [295]. Port expansion to accommodate these sectors intensifies land use conflicts and environmental pressures [296]. Blue port initiatives emphasize decarbonization and circular economy but often neglect social dimensions [297].

5.4 Shipping: Global Labor Regimes and Regulatory Gaps

Globalized labor markets: Shipping exemplifies globalized labor regimes characterized by international crewing, flags of convenience, and regulatory arbitrage [298], [299]. Seafarers are recruited globally, with distinct national and racial hierarchies: officers predominantly from developed countries, ratings from developing nations, particularly the Philippines, Indonesia, and Eastern Europe [300], [301].

This segmentation produces systematic wage differentials and unequal career opportunities [302]. Filipino seafarers, despite comprising a large share of global maritime labor, face lower wages and limited advancement compared to European officers [303]. Racialized hierarchies are reinforced through recruitment systems, training regimes, and shipboard social organization [304].

Precarity and vulnerability: Seafarers face multiple vulnerabilities: long periods away from home and family, social isolation, confined living conditions, and limited labor rights [305], [306]. Flags of convenience enable shipowners to avoid labor protections, safety standards, and union recognition [307]. Seafarers on substandard vessels face heightened risks of accidents, illness, and abandonment [308].

The COVID-19 pandemic starkly revealed seafarer vulnerabilities, with hundreds of thousands stranded at sea beyond contract periods due to travel restrictions and crew change difficulties [309]. Despite being designated "key workers," seafarers received minimal support, highlighting their marginalization in global governance [310].

Organizational dynamics and safety: Research on shipboard organization reveals how profit pressures, cost-cutting, and regulatory gaps systematically undermine safety [311], [312]. Reduced crew sizes, deferred maintenance, and inadequate training increase accident risks [313]. Multinational crews face communication challenges and cultural tensions that can compromise safety [314].

Organizational cultures emphasizing productivity over safety, combined with weak enforcement of regulations, produce predictable patterns of occupational injury and mortality [315]. Political economy analysis reveals how these outcomes reflect capitalist imperatives to externalize costs and maximize profits [316].

Collective organization challenges: Seafarers face structural barriers to collective organization: spatial dispersion across vessels, multinational composition hindering solidarity, employer strategies to fragment organization, and weak enforcement of union rights [317], [318]. International Transport Workers' Federation campaigns for flags of convenience abolition and decent work standards, but union density remains low [319].

Blue Economy connections: Shipping is essential infrastructure for Blue Economy, transporting goods, servicing offshore installations, and supporting marine industries [320]. Decarbonization initiatives (alternative fuels, slow steaming, port electrification) will transform shipping operations with implications for employment and skills [321]. Just transition frameworks must address how decarbonization affects maritime workers [322].

5.5 Offshore Energy: Social Impacts and Community Engagement

Emerging industry dynamics: Offshore wind, wave, and tidal energy represent rapidly expanding Blue Economy sectors [323]. These industries create new employment opportunities but also produce social impacts and conflicts [324]. Research reveals that social impacts often precede perceived benefits, creating community opposition if not addressed proactively [325].

Distributional conflicts: Offshore energy development raises questions of who benefits and who bears costs [326]. Energy companies and investors capture profits, while coastal communities experience visual impacts, fishing ground displacement, and tourism effects [327]. Benefit-sharing mechanisms are often weak or absent, producing distributional injustice [328].

Spatial conflicts emerge as offshore energy installations compete with fishing, shipping, recreation, and conservation for marine space [329]. Marine spatial planning processes that allocate areas to energy development without meaningful community participation produce legitimacy deficits [330].

Labor and skills: Offshore energy industries require specialized skills in engineering, construction, and maintenance [331]. Employment opportunities may not benefit local coastal communities if skills requirements exclude local workers [332]. Just transition frameworks emphasize ensuring that workers from declining industries (e.g., offshore oil and gas) can transition to renewable energy employment [333].

Working conditions in offshore energy involve extended periods on platforms, physically demanding work, and safety risks [334]. Labor standards and collective organization vary by national context and company practices [335].

Community engagement and legitimacy: Research demonstrates that early, sustained, and meaningful community engagement is critical for social acceptance of offshore energy projects [336], [337]. Procedural justice—fair, transparent, and inclusive decision-making processes—shapes community responses as much as distributional outcomes [338].

Projects that involve communities in planning, provide transparent information, establish benefit-sharing mechanisms, and create grievance redress procedures achieve greater legitimacy [339]. Conversely, top-down development that treats communities as obstacles to overcome produces opposition and conflict [340].

Cultural and place attachments: Offshore energy development affects cultural meanings and place attachments [341]. Fishing communities may view offshore installations as threats to maritime heritage and identity [342]. Coastal residents value seascapes for aesthetic, recreational, and spiritual reasons that are disrupted by industrial development [343]. Recognitional justice requires acknowledging and respecting these cultural dimensions [344].

5.6 Coastal Tourism: Uneven Development and Precarious Work

Tourism expansion and transformation: Coastal and marine tourism is a major Blue Economy sector, encompassing beach tourism, cruise ships, recreational boating, diving, and ecotourism [345]. Tourism development transforms coastal economies, landscapes, and social relations [346].

Uneven distribution of benefits: Tourism benefits are unevenly distributed [347]. Large tourism operators, hotel chains, and cruise companies capture substantial profits, while local communities receive limited benefits through low-wage service employment [348]. Tourism development often produces gentrification, displacing local residents and traditional livelihoods [349].

Cruise tourism exemplifies these dynamics [350]. Cruise ships bring thousands of visitors to ports but generate minimal local economic benefits as tourists spend primarily onboard and at company-controlled shore excursions [351]. Ports bear infrastructure costs and environmental impacts while capturing little revenue [352].

Precarious labor: Tourism employment is characterized by seasonality, low wages, limited benefits, and weak collective organization [353]. Workers in hotels, restaurants, and recreational services face precarious conditions with minimal job security [354]. Gender dimensions are significant, with women concentrated in lower-paid service roles [355].

Environmental and social impacts: Mass tourism produces environmental degradation—coastal development, water consumption, waste generation, ecosystem damage—that undermines the natural amenities attracting tourists [356]. Social impacts include cultural commodification, loss of community cohesion, and conflicts between residents and tourists [357].

Sustainable tourism alternatives: Sustainable and community-based tourism models offer alternatives emphasizing local ownership, environmental stewardship, and cultural respect [358]. However, these remain niche compared to mass tourism [359]. Blue Economy frameworks often promote tourism expansion without adequate attention to sustainability and equity [360].

5.7 Comparative Synthesis: Common Patterns and Sectoral Specificities

Across sectors, common patterns emerge:

Labor stratification: All sectors exhibit hierarchies based on capital ownership, skill levels, nationality, race, and gender [361]. Industrial and capital-intensive operations dominate over small-scale and artisanal activities [362]. Precarious employment, weak labor protections, and limited collective organization characterize much Blue Economy work [363].

Power asymmetries: Large corporations, state agencies, and international organizations wield disproportionate power in shaping sectoral development and governance [364]. Small-scale users, workers, and coastal communities have limited influence despite being most affected by Blue Economy transformations [365].

Participation deficits: Governance processes across sectors privilege technical expertise and sectoral representation over democratic participation and community empowerment [366]. Decisions are made through technocratic processes that marginalize affected communities [367].

Distributional conflicts: Blue Economy development produces winners and losers, with benefits concentrated among capital owners and costs borne by workers, small-scale users, and coastal communities [368]. Benefit-sharing mechanisms are weak or absent [369].

Spatial competition: Expanding multiple ocean uses simultaneously creates spatial conflicts and cumulative impacts [370]. Marine spatial planning attempts to coordinate uses but often reproduces power asymmetries and marginalizes small-scale users [371].

Yet sectoral specificities matter:

- Fisheries involve direct dependence on marine ecosystems and traditional livelihoods, making justice and rights frameworks particularly salient [372]
- Ports are sites of intensive territorial transformation and labor restructuring, highlighting maritimization dynamics [373]
- Shipping exemplifies globalized labor regimes and regulatory arbitrage, foregrounding political economy of flags of convenience [374]
- Offshore energy raises novel questions of marine industrialization, spatial allocation, and community engagement [375]
- Tourism involves complex interactions between economic development, environmental impacts, and cultural transformation [376]

Blue Sociology must attend to both common patterns and sectoral specificities, developing analytical frameworks flexible enough to address diverse contexts while identifying systemic dynamics across sectors.

---

6. Research Gaps and Social Science Solutions {#6-research-gaps-and-social-science-solutions}

6.1 Introduction: Systematic Deficits in Blue Economy Research

Comprehensive literature synthesis reveals recurring conceptual and methodological gaps in Blue Economy research and policy [377]. These gaps are not random but reflect systematic biases: privileging economic over social dimensions, natural science over social science, quantitative metrics over qualitative understanding, technocratic expertise over experiential knowledge, and growth imperatives over justice concerns [378].

This section systematically identifies seven major research gaps and demonstrates how maritime social science methods can address each deficit. The analysis draws on the blue economy gaps synthesis to specify: (1) gap description, (2) how social science can help, (3) specific methodological potential, (4) why these approaches work, and (5) key supporting evidence.

6.2 Gap 1: Equity and Justice Marginalization

Gap description: Blue Economy policies and analyses often treat economic growth as value-neutral, sidelining distributive and recognitional justice for small-scale fishers and coastal communities [379], [380]. This enables dispossession, marginalization, and exclusion from decision-making processes [381].

How social science helps: Political ecology and blue justice frameworks surface historical, economic, and institutional drivers of inequity [382]. These approaches reframe analyses to center rights, livelihoods, and food security rather than economic efficiency [383].

Specific methodological potential:

- Political ecology frameworks linking macro-level political economy with local experiences
- Blue justice approaches analyzing distributive, procedural, and recognitional justice dimensions
- Rights-based frameworks emphasizing human rights, community rights, and rights of nature
- Historical analysis revealing how contemporary inequalities emerged from past policies

Why this works: These frameworks make visible processes—industrialization, spatial enclosure, regulatory capture—that produce inequity [384]. By linking structural drivers to lived experiences, they inform redistributive and recognition-oriented policy interventions [385]. Political ecology's attention to power relations reveals whose interests are served by ostensibly neutral policies [386].

Evidence base: Research demonstrates that blue economy industrialization systematically marginalizes small-scale fisheries through quota allocation, spatial planning, and market concentration [387], [388]. Political ecology analysis reveals how these outcomes reflect capitalist accumulation dynamics and state priorities favoring industrial development [389]. Blue justice frameworks provide normative foundations for alternative policies centered on equity [390].

6.3 Gap 2: Participation and Inclusion Deficits

Gap description: Rapid blue growth promotion frequently fast-tracks planning and investment while marginalizing affected communities, producing conflicts and weak social legitimacy [391], [392]. Consultation processes, when they occur, are

often superficial and lack genuine power-sharing [393].

How social science helps: Deliberative, participatory, and co-production approaches from maritime social science open inclusive spaces for knowledge exchange and collective decision-making [394]. These methods shift processes away from top-down rollouts toward collaborative governance [395].

Specific methodological potential:

- Participatory action research involving communities as co-researchers
- Co-production of knowledge integrating scientific, local, and traditional knowledge
- Stakeholder-driven scenario building exploring alternative futures
- Deliberative forums enabling structured dialogue across diverse actors
- Community-based monitoring and adaptive management

Why this works: Participatory methods build trust, surface local priorities and values, and produce actionable local buy-in [396]. They reveal trade-offs and alternatives early, reducing later conflict and distributional harms [397]. Co-production recognizes diverse knowledge systems as legitimate, challenging technocratic monopolies on expertise [398].

Evidence base: Studies document how participatory approaches in marine spatial planning, fisheries co-management, and marine protected area design improve both social outcomes and ecological effectiveness [399], [400]. Conversely, top-down processes produce opposition, non-compliance, and legitimacy crises [401]. Participatory methods are particularly effective when they involve genuine power-sharing rather than tokenistic consultation [402].

6.4 Gap 3: Governance Fragmentation and Power Asymmetries

Gap description: Governance arrangements across sectors and scales are fragmented [403]. Power asymmetries favor industry and state actors, constraining small-scale users and producing exclusion and ocean-grabbing dynamics [404], [405].

How social science helps: Multi-level governance analysis, political ecology, and place-based research highlight where institutional fragmentation and power concentrations undermine equitable outcomes [406]. These approaches identify loci for governance reform [407].

Specific methodological potential:

- Multi-scalar governance analysis tracing institutions and power across local, national, regional, and global levels
- Political ecology frameworks analyzing how governance arrangements reflect and reproduce power relations
- Place-based participatory mapping revealing local knowledge and spatial claims
- Institutional analysis examining formal and informal rules shaping resource access
- Network analysis mapping actor relationships and influence

Why this works: These methods trace institutions and power relations across scales, enabling targeted governance reforms [408]. They reveal how fragmentation serves particular interests by obscuring accountability and enabling regulatory arbitrage [409]. Place-based approaches ground abstract governance analysis in lived experiences and local contexts [410].

Evidence base: Research on marine governance in regions like the Coral Triangle demonstrates how institutional fragmentation and power asymmetries undermine sustainability and equity [411]. Multi-level governance analysis reveals scale mismatches and coordination failures [412]. Political ecology studies show how governance reforms (co-management, fiscal measures, rights recognition) can reallocate authority and benefits toward marginalized actors when backed by political will [413].

6.5 Gap 4: Methodological Bias Toward Economic Metrics

Gap description: Economic indicators dominate assessments of blue initiatives, underrepresenting wellbeing, food security, cultural values, and cumulative impacts on vulnerable communities [414], [415]. GDP and sectoral growth metrics crowd out social and ecological indicators [416].

How social science helps: Social-ecological systems theory, mixed-methods evaluation, and social impact assessment incorporate non-market values, distributional outcomes, and long-term resilience into appraisal frameworks [417], [418].

Specific methodological potential:

- Social-ecological systems theory linking social and ecological dynamics
- Mixed-methods evaluation combining quantitative and qualitative data
- Participatory indicators developed with communities
- Social impact assessment examining distributional effects
- Wellbeing frameworks measuring quality of life beyond income
- Cultural ecosystem services valuation

Why this works: SES theory links social and ecological dynamics, supporting indicators that reflect ecosystem services, livelihoods, and equity rather than only monetary outputs [419]. This enables more holistic and precautionary policy choices

[420]. Mixed methods capture both breadth (quantitative patterns) and depth (qualitative meanings and experiences) [421].

Evidence base: Studies demonstrate that economic metrics alone provide misleading assessments of blue economy impacts [422]. Communities may experience declining wellbeing despite GDP growth if benefits are captured by external actors and costs are borne locally [423]. Social-ecological systems approaches reveal trade-offs and synergies invisible in purely economic analysis [424].

6.6 Gap 5: Indigenous and Local Knowledge Exclusion

Gap description: Planning rarely integrates Indigenous and local knowledge or examines gendered roles and impacts [425], [426]. This results in lost stewardship practices and disproportionate burdens on women and marginalized groups [427].

How social science helps: Ethnography, feminist political ecology, and Indigenous knowledge integration enable documentation of local practices, gendered labor, and place-based governance essential for equitable and adaptive blue policies [428], [429].

Specific methodological potential:

- Ethnographic methods documenting lived experiences and local practices
- Feminist political ecology analyzing gendered dimensions of resource use and governance
- Indigenous knowledge integration respecting Indigenous epistemologies and protocols
- Participatory mapping revealing spatial knowledge and customary tenure
- Oral history capturing intergenerational knowledge transmission
- Gender analysis examining differential impacts and opportunities

Why this works: These approaches surface lived experience, gendered divisions of labor, and non-market values [430]. They can be operationalized in co-management or rights-based policies that protect cultural practices and ensure benefit sharing [431]. Ethnography provides depth and context that surveys and quantitative methods miss [432].

Evidence base: Research documents how Indigenous and local knowledge systems embody sophisticated understanding of marine ecosystems and sustainable use practices developed over generations [433], [434]. Excluding this knowledge produces management failures and social injustice [435]. Feminist analysis reveals how gender-blind policies reproduce inequalities and miss opportunities for more equitable outcomes [436].

6.7 Gap 6: Power Asymmetries and Social License Risks

Gap description: Projects may achieve regulatory approval while lacking social license to operate, reflecting unequal negotiation power and limited mechanisms to address community grievances [437], [438]. This produces conflicts, protests, and project delays or failures [439].

How social science helps: Critical studies of social license, advocacy research, and conflict analysis expose asymmetries in negotiation [440]. These approaches enable design of governance mechanisms that mandate consent, transparency, and accountability [441].

Specific methodological potential:

- Social license to operate analysis examining legitimacy dimensions
- Conflict mapping identifying stakeholders, interests, and power relations
- Advocacy research supporting community organizing and rights claims
- Legal empowerment research strengthening community capacity to engage governance
- Grievance mechanism assessment evaluating accessibility and effectiveness
- Corporate accountability research documenting impacts and responsibilities

Why this works: SLO research documents legitimacy gaps and offers procedural remedies—early engagement, transparent benefit sharing, grievance redress—that reduce contestation and protect vulnerable stakeholders [442]. Conflict analysis reveals underlying interests and potential resolution pathways [443]. Advocacy research explicitly supports marginalized actors in asymmetric power relations [444].

Evidence base: Studies show that projects lacking social license face opposition, delays, and failures despite regulatory approval [445]. Conversely, projects that invest in building legitimacy through fair processes and benefit-sharing achieve smoother implementation [446]. Power asymmetries mean that procedural fairness alone is insufficient; substantive redistribution of benefits and decision-making authority is necessary [447].

6.8 Gap 7: Social-Ecological Integration Deficits

Gap description: Blue Economy frameworks insufficiently integrate ecological dynamics with social processes across scales [448]. Social and ecological dimensions are analyzed separately rather than as coupled systems [449]. This produces management failures when social-ecological feedbacks are ignored [450].

How social science helps: Social-ecological systems theory, political ecology, and environmental sociology provide frameworks for analyzing coupled human-natural systems [451]. These approaches examine feedbacks, thresholds, and resilience in integrated ways [452].

Specific methodological potential:

- Social-ecological systems theory analyzing coupled dynamics
- Political ecology examining how power relations shape environmental conditions
- Environmental sociology studying social metabolism and ecological distribution conflicts
- Resilience thinking analyzing adaptive capacity and transformation
- Participatory modeling engaging stakeholders in systems analysis
- Integrated assessment combining social and ecological data

Why this works: Social-ecological approaches recognize that human and natural systems are fundamentally interconnected [453]. Management interventions produce social-ecological outcomes that cannot be predicted by analyzing social or ecological dimensions separately [454]. Integrated frameworks enable anticipating feedbacks, identifying leverage points, and designing interventions that address root causes [455].

Evidence base: Research demonstrates that fisheries management, marine conservation, and coastal development produce social-ecological outcomes shaped by complex feedbacks [456]. Ignoring social dimensions produces ecological failures (e.g., non-compliance with regulations) [457]. Ignoring ecological dimensions produces social failures (e.g., livelihood collapse from resource depletion) [458]. Integrated social-ecological analysis improves both social and ecological outcomes [459].

6.9 Synthesis: Toward Methodologically Pluralist Blue Sociology

These seven gaps reveal systematic deficits in how Blue Economy is researched and governed. Addressing them requires methodological pluralism that integrates:

- Critical frameworks (political ecology, feminist analysis, decolonial approaches) that foreground power, inequality, and justice
- Participatory methods (action research, co-production, deliberative forums) that democratize knowledge production and decision-making
- Integrative approaches (social-ecological systems, multi-level governance, mixed methods) that address complexity and cross-scale dynamics

- Qualitative depth (ethnography, oral history, case studies) that captures meanings, experiences, and contexts
- Quantitative breadth (surveys, indicators, spatial analysis) that identifies patterns and scales up insights

Blue Sociology provides the conceptual architecture for this methodological pluralism, positioning social science not as auxiliary to natural science but as equally essential for understanding and governing sea societies. The next section elaborates this analytical framework.

---

7. The Blue Sociology Framework: Analytical Architecture {#7-the-blue-sociology-framework}

7.1 Defining Blue Sociology: Complementary Renewal

Blue Sociology is proposed as a complementary renewal and extension of maritime sociology toward an integrated sociology of sea societies in the context of accelerating Blue Economy development. It is not a replacement for maritime sociology but builds on its empirical strengths and analytical traditions while incorporating theoretical innovations from the oceanic turn, environmental justice scholarship, and critical governance studies.

Blue Sociology responds to three specific needs:

1. Integrated cross-sectoral analysis: Blue Economy governance increasingly involves complex interactions across fisheries, shipping, energy, tourism, and conservation. Maritime sociology's traditional sectoral focus must be extended to address these cross-sectoral dynamics and cumulative impacts.

2. Socioecological integration: Contemporary ocean challenges are fundamentally social-ecological, involving coupled human-natural systems, multispecies entanglements, and planetary-scale processes. Maritime sociology must integrate environmental sociology and oceanic turn insights to address these dimensions.

3. Justice and transformation: Blue Economy development raises urgent questions of equity, power, and democratic governance. Maritime sociology must engage more deeply with environmental justice frameworks and transformative change scholarship to address these normative and political dimensions.

Blue Sociology synthesizes maritime sociology's labor-centered, institutionally grounded, and historically informed approaches with oceanic turn's relational, materialist, and more-than-human perspectives. This synthesis enables analysis that is simultaneously attentive to: labor regimes and working conditions, institutional arrangements and power relations, multispecies entanglements and material agencies, spatial dynamics and territorial transformations, governance mechanisms and democratic legitimacy, and justice dimensions across distribution, procedure, and recognition.

7.2 The Tripartite Framework: Marine-Maritime-Oceanic

Blue Sociology proposes a tripartite analytical framework distinguishing three interconnected domains:

Marine Domain: Natural oceanographic and ecological processes

- Physical oceanography: currents, tides, temperature, salinity, chemistry

- Marine ecology: species, populations, communities, ecosystems, food webs
- Biogeochemical cycles: carbon, nitrogen, oxygen, nutrient flows
- Geological processes: seafloor formation, sedimentation, tectonics
- Climate interactions: ocean-atmosphere exchanges, heat storage, circulation

The marine domain encompasses natural processes that exist independently of human activity (though increasingly affected by it). Understanding this domain requires natural science—oceanography, marine biology, ecology, geochemistry. However, Blue Sociology recognizes that "natural" processes are increasingly shaped by human activities (climate change, pollution, species introductions) and that human-ocean relations are mediated by how marine processes are known, represented, and governed.

Maritime Domain: Human activities, labor regimes, and institutional arrangements at sea

- Maritime industries: fishing, shipping, offshore energy, aquaculture, tourism
- Labor and work: occupational communities, employment relations, working conditions
- Organizations: firms, unions, cooperatives, associations
- Technologies: vessels, gear, platforms, infrastructure
- Institutions: property rights, regulations, governance arrangements
- Territorial dynamics: ports, coastal zones, maritime boundaries
- Cultural practices: maritime heritage, seafaring traditions, coastal identities

The maritime domain encompasses human activities in and relating to marine spaces. This is maritime sociology's traditional focus—analyzing work, organizations, institutions, and communities involved in ocean use. Blue Sociology extends this through: cross-sectoral integration addressing interactions among maritime industries, labor-centered analysis foregrounding working conditions and power relations, institutional analysis examining how rules and norms shape access and outcomes, territorial analysis of how maritime activities reshape coastal spaces, and cultural analysis of meanings, identities, and knowledge systems.

Oceanic Domain: Planetary-scale socioecological systems and governance challenges

- Global ocean systems: circulation patterns, climate regulation, biogeochemical cycles
- Planetary boundaries: ocean acidification, biodiversity loss, pollution
- Transnational governance: international law, regional agreements, global institutions
- Scalar interactions: how local, national, regional, and global processes interconnect
- Socioecological metabolism: material and energy flows between society and oceans
- Multispecies assemblages: entanglements of human and nonhuman actors across scales
- Futures and transformations: trajectories, tipping points, alternative pathways

The oceanic domain addresses planetary-scale socioecological systems and governance challenges. This is where Blue Sociology most distinctly extends maritime sociology, incorporating: oceanic turn insights on relational, fluid, and more-than-human ocean spaces, environmental sociology frameworks on social metabolism and ecological distribution, global governance analysis of transnational institutions and scalar politics, Earth system science perspectives on planetary boundaries and tipping points, and futures thinking about transformation pathways and alternative ocean futures.

Interconnections: These three domains are analytically distinct but empirically interconnected. Marine processes shape maritime activities (e.g., fish migration patterns affect fishing practices). Maritime activities affect marine processes (e.g., fishing pressure alters ecosystem structure). Both are embedded in oceanic-scale systems (e.g., global seafood trade connects local fisheries to planetary commodity chains). Governance operates across all three domains (e.g., climate policy affects ocean chemistry, which affects fisheries, which affects coastal livelihoods).

Blue Sociology's analytical power lies in examining these interconnections—how marine, maritime, and oceanic domains co-constitute contemporary sea societies. This tripartite framework enables:

- Tracing how planetary processes (climate change, ocean acidification) affect local maritime communities
- Analyzing how local maritime practices (fishing, shipping) aggregate to oceanic-scale impacts
- Examining how governance institutions mediate between marine processes, maritime activities, and oceanic systems
- Identifying leverage points for intervention across domains and scales

7.3 Core Analytical Dimensions

Blue Sociology organizes analysis around six core dimensions that cut across the marine-maritime-oceanic framework:

1. Labor and Social Reproduction

- Work processes, skills, and occupational identities in maritime industries
- Employment relations, labor markets, and stratification

- Working conditions, safety, and occupational health
- Social reproduction: how maritime work affects families and communities
- Gender divisions of labor and care work
- Collective organization, unions, and worker power
- Just transitions in changing maritime industries

This dimension centers labor—a defining focus of maritime sociology—while extending analysis to include social reproduction, gender, and just transitions. It examines how Blue Economy development affects workers, families, and communities, and how labor relations shape social and ecological outcomes.

2. Institutional Arrangements and Power Relations

- Property rights and access regimes to marine resources and spaces
- Regulatory frameworks and enforcement mechanisms
- Governance institutions at multiple scales
- Power asymmetries among actors (states, corporations, communities)
- Political economy of maritime industries
- Lobbying, influence, and regulatory capture
- Resistance, contestation, and social movements

This dimension analyzes institutions—formal and informal rules—that structure ocean use and governance. It foregrounds power relations, examining who has authority to make decisions, whose interests are served by institutional arrangements, and how power is exercised, contested, and potentially transformed.

3. Spatial Dynamics and Territorial Transformations

- Maritimization: how maritime activities reshape territories
- Port development and urban-ocean interfaces
- Marine spatial planning and spatial allocation
- Spatial conflicts and cumulative impacts
- Coastal gentrification and displacement
- Maritime boundaries and jurisdictional politics
- Spatial justice and equitable access to marine spaces

This dimension examines spatial aspects of Blue Economy development, drawing on maritime sociology's territorial analysis and human geography's spatial justice frameworks. It analyzes how marine spaces are produced, governed, and contested, and how spatial arrangements reflect and reproduce social inequalities.

4. Socioecological Relations and Multispecies Entanglements

- Human-ocean relations and dependencies
- Multispecies interactions in fisheries, aquaculture, and conservation
- Material agencies of water, weather, and marine life
- Ecosystem services and social metabolism
- Pollution, degradation, and ecological distribution conflicts
- Climate change impacts on marine systems and maritime communities
- More-than-human perspectives on ocean governance

This dimension integrates environmental sociology and oceanic turn insights, analyzing coupled social-ecological systems. It examines how human activities affect marine ecosystems and how marine processes shape human societies, attending to multispecies entanglements and material agencies often absent from conventional social science.

5. Knowledge, Culture, and Meaning

- Scientific knowledge production and epistemic authority
- Traditional and Indigenous knowledge systems
- Local and experiential knowledge of maritime communities
- Cultural meanings of oceans, seas, and coasts
- Maritime heritage and seafaring traditions

- Representations and imaginaries of oceans
- Knowledge politics in ocean governance

This dimension examines how oceans are known, represented, and made meaningful. It analyzes knowledge politics—whose knowledge counts in governance—and cultural dimensions of human-ocean relations. It challenges epistemic hierarchies that privilege Western science over other knowledge systems.

6. Justice, Democracy, and Transformation

- Distributive justice: equitable sharing of benefits and burdens
- Procedural justice: fair, inclusive, and transparent decision-making
- Recognitional justice: respect for diverse identities, knowledge, and values
- Democratic participation and community empowerment
- Rights-based frameworks (human rights, community rights, rights of nature)
- Social movements and struggles for blue justice
- Transformation pathways toward equitable and sustainable ocean futures

This dimension addresses normative and political questions central to Blue Economy debates. It analyzes justice dimensions, democratic governance, and pathways for transformative change. It positions Blue Sociology not just as analytical but as engaged with struggles for more just and sustainable ocean futures.

7.4 Methodological Principles

Blue Sociology is characterized by methodological pluralism guided by several principles:

Multi-scalar analysis: Examining processes and institutions across local, regional, national, and global scales, and analyzing how scales interact and shape each other.

Historical depth: Situating contemporary developments in historical context, tracing path dependencies and transformative moments, learning from past experiences.

Comparative perspective: Analyzing similarities and differences across sectors, regions, and governance systems to identify patterns and context-specific factors.

Integrative approaches: Combining social and ecological analysis, quantitative and qualitative methods, and diverse disciplinary perspectives.

Participatory orientation: Engaging affected communities as knowledge producers and decision-makers, not just research subjects or stakeholders.

Critical reflexivity: Examining how research itself is positioned within power relations, whose interests it serves, and how it might contribute to or challenge inequalities.

Transdisciplinary collaboration: Working across natural sciences, social sciences, humanities, and with practitioners and communities to address complex problems.

Justice commitment: Explicitly engaging with normative questions and positioning research to support more equitable and democratic ocean governance.

7.5 Relationship to Maritime Sociology and Oceanic Turn

Blue Sociology is positioned as complementary renewal rather than replacement:

Continuity with maritime sociology:

- Maintains focus on labor, work, and occupational communities
- Continues institutional and organizational analysis
- Preserves political economy critique of maritime industries
- Builds on territorial analysis of maritimization
- Extends historical and comparative traditions

Extension beyond maritime sociology:

- Expands from sectoral focus to cross-sectoral integration
- Incorporates socioecological and more-than-human perspectives
- Engages planetary-scale processes and governance
- Deepens justice and democratic governance analysis

- Integrates oceanic turn theoretical innovations

Integration with oceanic turn:

- Adopts relational, materialist, and more-than-human ontologies
- Incorporates hydrosocial and infrastructural analysis
- Engages Indigenous and decolonial ocean scholarship
- Applies feminist and intersectional perspectives
- Embraces methodological innovations attuned to aquatic specificities

Distinctive contribution: Blue Sociology's distinctive contribution is synthesizing maritime sociology's empirical depth and labor-centered analysis with oceanic turn's theoretical innovations and justice commitments. This synthesis enables analysis that is simultaneously:

- Grounded in material conditions of maritime work and coastal livelihoods
- Attentive to multispecies entanglements and material agencies
- Critical of power relations and political economy
- Engaged with justice and democratic transformation
- Integrated across social-ecological systems and scales

7.6 Applications and Research Domains

Blue Sociology frameworks can be applied to diverse research domains:

Sectoral studies: Analyzing specific Blue Economy sectors (fisheries, shipping, offshore energy, tourism, aquaculture) through integrated social-ecological and justice lenses.

Governance analysis: Examining ocean governance institutions, policies, and processes at multiple scales, assessing effectiveness, equity, and legitimacy.

Labor studies: Investigating working conditions, employment relations, and labor organizing across maritime industries, with attention to gender, race, and migration.

Spatial analysis: Studying marine spatial planning, territorial transformations, and spatial justice in ocean use and governance.

Impact assessment: Evaluating social, economic, cultural, and ecological impacts of Blue Economy development, with emphasis on distributional effects.

Participatory research: Engaging communities in co-producing knowledge, designing governance alternatives, and building capacity for collective action.

Transformation studies: Analyzing pathways, barriers, and strategies for transforming toward more equitable and sustainable ocean futures.

Comparative research: Examining similarities and differences across regions, sectors, and governance systems to identify transferable lessons and context-specific factors.

The next section elaborates methodological implications and proposes a research agenda for Blue Sociology.

---

8. Methodological Implications and Research Agenda {#8-methodological-implications}

8.1 Methodological Pluralism in Practice

Blue Sociology requires methodological pluralism that integrates diverse approaches suited to different analytical dimensions and research questions [460]. This section outlines key methodological approaches and their applications.

Ethnographic methods: Participant observation, in-depth interviews, and extended fieldwork enable deep understanding of lived experiences, cultural meanings, and social practices in maritime communities [461]. Ethnography is particularly valuable for: documenting working conditions and labor processes aboard vessels and in ports, understanding how maritime communities experience and respond to Blue Economy changes, revealing informal institutions and practices not captured in formal analysis, and analyzing cultural dimensions of human-ocean relations [462].

Political economy analysis: Examining ownership structures, profit flows, market dynamics, and regulatory regimes reveals how capitalist dynamics shape maritime industries [463]. Political economy approaches are essential for: tracing how global commodity chains connect local fisheries to international markets, analyzing how flags of convenience and regulatory arbitrage produce labor precarity, examining corporate concentration and market power in maritime sectors, and understanding how financialization affects ocean governance [464].

Institutional analysis: Studying formal and informal rules, governance arrangements, and organizational structures illuminates how institutions shape access, behavior, and outcomes [465]. Institutional methods include: analyzing property rights regimes and access rules to marine resources, examining regulatory frameworks and enforcement mechanisms, studying governance institutions at multiple scales and their interactions, and assessing institutional change and reform processes [466].

Participatory action research: Engaging communities as co-researchers and co-designers democratizes knowledge production and supports collective action [467]. PAR approaches involve: collaborative problem definition with affected communities, participatory data collection and analysis, co-design of governance alternatives and interventions, and capacity building for community organizing and advocacy [468].

Social-ecological systems analysis: Integrating social and ecological data and methods enables analysis of coupled human-natural systems [469]. SES approaches include: participatory modeling of social-ecological dynamics, resilience assessment examining adaptive capacity and thresholds, scenario analysis exploring alternative futures, and integrated monitoring combining social and ecological indicators [470].

Spatial analysis and mapping: Geographic methods reveal spatial patterns, conflicts, and justice dimensions [471]. Spatial approaches include: participatory mapping documenting local knowledge and spatial claims, GIS analysis of marine spatial planning and cumulative impacts, spatial justice assessment examining distributional equity, and network analysis mapping actor relationships and flows [472].

Historical analysis: Examining long-term trajectories, path dependencies, and transformative moments provides context for contemporary developments [473]. Historical methods include: archival research on maritime industries and governance, oral history capturing intergenerational knowledge and change, comparative historical analysis identifying patterns and contingencies, and longue durée analysis of sociotechnical trajectories [474].

Mixed methods integration: Combining quantitative and qualitative approaches enables both breadth and depth [475]. Mixed methods designs include: sequential designs where qualitative work informs quantitative instruments or vice versa, concurrent designs collecting both types of data simultaneously, and embedded designs where one method is nested within another [476].

8.2 Research Agenda: Priority Themes

Blue Sociology research agenda encompasses multiple priority themes:

Theme 1: Labor regimes and just transitions across Blue Economy sectors

Research questions:

- How do labor conditions, employment relations, and worker power vary across maritime sectors?
- What are the impacts of automation, digitalization, and decarbonization on maritime employment?
- How can just transition frameworks ensure that workers benefit from Blue Economy transformations?
- What strategies enable effective labor organizing in globalized maritime industries?
- How do gender, race, nationality, and migration status shape maritime labor stratification?

Methods: Ethnography of work processes, labor market analysis, comparative case studies, participatory action research with worker organizations, policy analysis of just transition initiatives.

Theme 2: Governance institutions and democratic legitimacy in ocean governance

Research questions:

- How do governance institutions at multiple scales interact and shape ocean use?
- What mechanisms enable meaningful community participation in marine spatial planning?
- How can governance address power asymmetries between industrial and small-scale users?
- What institutional reforms would enhance accountability and transparency in ocean governance?
- How do different governance models (state-led, market-based, community-based, hybrid) perform on equity and sustainability criteria?

Methods: Institutional analysis, multi-level governance assessment, participatory governance experiments, comparative case studies, deliberative forums, policy evaluation.

Theme 3: Spatial justice and territorial transformations in coastal regions

Research questions:

- How does Blue Economy development reshape coastal territories and urban-ocean interfaces?
- What are the distributional effects of marine spatial planning across different user groups?
- How do port expansion and maritime industrialization affect coastal communities?
- What mechanisms can ensure spatial justice in ocean use allocation?
- How do coastal gentrification and displacement processes unfold in Blue Economy contexts?

Methods: Spatial analysis and GIS, participatory mapping, ethnography of coastal communities, political ecology of territorial change, comparative urban analysis, spatial justice assessment.

Theme 4: Social-ecological dynamics and multispecies relations in marine systems

Research questions:

- How do social and ecological processes interact in fisheries, aquaculture, and conservation?
- What role do nonhuman actors and material agencies play in shaping ocean governance outcomes?
- How do climate change impacts on marine ecosystems affect maritime communities?
- What are the social-ecological feedbacks in Blue Economy development?
- How can governance address coupled social-ecological dynamics?

Methods: Social-ecological systems analysis, participatory modeling, resilience assessment, more-than-human ethnography, integrated monitoring, scenario analysis.

Theme 5: Knowledge politics and epistemic justice in ocean science and governance

Research questions:

- Whose knowledge counts in ocean governance and why?
- How can Indigenous and local knowledge be respectfully integrated with scientific knowledge?
- What are the power dynamics in ocean science and knowledge production?
- How do representations and imaginaries of oceans shape governance possibilities?
- What would epistemic justice look like in ocean governance?

Methods: Discourse analysis, ethnography of science, Indigenous knowledge integration, participatory knowledge co-production, comparative epistemology, knowledge politics analysis.

Theme 6: Blue justice movements and struggles for equitable ocean futures

Research questions:

- How do social movements challenge blue economy industrialization and advocate for alternatives?
- What strategies enable effective organizing by small-scale fishers and coastal communities?
- How do justice claims (distributive, procedural, recognitional) manifest in ocean conflicts?
- What alliances and solidarities emerge across maritime sectors and regions?
- What alternative visions of ocean futures do movements articulate?

Methods: Social movement analysis, participatory action research, advocacy research, discourse analysis, comparative case studies, network analysis.

Theme 7: Transformation pathways toward sustainable and equitable ocean futures

Research questions:

- What are the barriers to and enablers of transformative change in ocean governance?
- How can Blue Economy transitions be designed to advance rather than undermine justice?
- What alternative paradigms (commons governance, degrowth, rights-based approaches) offer for ocean futures?
- What are the roles of different actors (states, corporations, communities, movements) in transformation?
- How can research support transformative change processes?

Methods: Transformation studies, scenario analysis, participatory futures visioning, comparative analysis of alternatives, action research, policy innovation experiments.

8.3 Capacity Building and Institutional Development

Advancing Blue Sociology requires capacity building and institutional development:

Training and education: Developing curricula, courses, and training programs in Blue Sociology for graduate students and early-career researchers. Integrating Blue Sociology perspectives into maritime studies, environmental sociology, and ocean governance programs.

Interdisciplinary collaboration: Building research networks and collaborations across maritime sociology, environmental sociology, human geography, political ecology, science and technology studies, and ocean sciences. Creating spaces for dialogue and knowledge exchange across disciplines.

Community engagement: Developing partnerships with fishing communities, maritime workers, coastal residents, and social movements. Supporting community-based research and participatory knowledge production. Ensuring research serves community needs and priorities.

Policy engagement: Translating research findings for policy audiences. Engaging with governance institutions at multiple scales. Supporting evidence-informed policy development while maintaining critical independence.

International networks: Building global networks of Blue Sociology researchers. Facilitating South-South and North-South knowledge exchange. Addressing colonial legacies in ocean research and governance.

Funding and resources: Advocating for research funding that supports social science in ocean contexts. Developing funding mechanisms that enable long-term, community-engaged research. Ensuring resources for participatory and action research approaches.

8.4 Ethical Considerations

Blue Sociology research must attend to ethical dimensions:

Positionality and reflexivity: Researchers must reflect on their own positions within power relations and how this shapes research. This includes examining privileges, biases, and assumptions, and being transparent about research motivations and funding sources.

Community consent and ownership: Research with communities requires free, prior, and informed consent. Communities should have ownership over data and findings, and control over how research is used and disseminated.

Do no harm: Research must not exacerbate vulnerabilities or expose communities to risks. This requires careful attention to confidentiality, security, and potential negative consequences of research.

Reciprocity and benefit-sharing: Research should provide tangible benefits to participating communities, not just extract knowledge. This may include capacity building, advocacy support, or resource sharing.

Epistemic justice: Research must respect diverse knowledge systems and avoid epistemic violence—imposing Western frameworks that delegitimize other ways of knowing.

Engaged scholarship: Blue Sociology is not neutral but engaged with struggles for justice and sustainability. This requires clarity about normative commitments while maintaining analytical rigor.

---

9. Policy Implications: Toward Just Blue Transitions {#9-policy-implications}

9.1 From Blue Growth to Blue Justice

Blue Sociology analysis reveals fundamental tensions between blue growth imperatives and social justice requirements [477]. Addressing these tensions requires reorienting Blue Economy policy from growth-centered to justice-centered frameworks [478]. This section outlines policy implications across governance, labor, participation, spatial planning, and knowledge integration.

9.2 Governance Reform: Democratization and Accountability

Multi-level governance coordination: Establish mechanisms for coordinating ocean governance across local, national, regional, and global scales while respecting subsidiarity [479]. Ensure that higher-level policies do not override local governance capacities and community rights [480].

Participatory governance institutions: Create spaces for meaningful community participation in ocean governance, not just consultation [481]. This includes: community representation in decision-making bodies with genuine voting power, participatory budgeting for ocean-related investments, deliberative forums for discussing values and priorities, and co-management arrangements sharing authority between state and communities [482].

Accountability mechanisms: Strengthen accountability through: transparent decision-making processes with public access to information, independent oversight and evaluation of social and environmental impacts, grievance mechanisms accessible to affected communities, and legal standing for communities to challenge decisions [483].

Rights-based frameworks: Adopt rights-based approaches recognizing: human rights to healthy ocean environment and sustainable livelihoods, community rights to marine resources and customary territories, Indigenous rights to self-determination and free, prior, and informed consent, and rights of nature or ocean ecosystems [484].

9.3 Labor Standards and Just Transitions

Decent work standards: Enforce labor rights and decent work standards across maritime sectors through: ratification and implementation of ILO maritime labor conventions, strengthened inspection and enforcement mechanisms, elimination of flags of convenience and regulatory arbitrage, and protection of rights to organize and bargain collectively [485].

Just transition frameworks: Design Blue Economy transitions to benefit workers through: retraining and skills development for workers in declining industries, income support during transitions, worker participation in transition planning, and ensuring new jobs are decent jobs with fair wages and conditions [486].

Gender equity: Address gender inequalities in maritime sectors through: equal pay and opportunity policies, recognition and compensation for women's contributions, gender-responsive governance and decision-making, and addressing harassment and discrimination [487].

Migrant worker protections: Strengthen protections for migrant maritime workers through: portable benefits and social protections, access to justice and grievance mechanisms, protection from exploitation and trafficking, and pathways to permanent residence and citizenship [488].

9.4 Spatial Justice in Marine Spatial Planning

Equitable spatial allocation: Ensure marine spatial planning addresses distributional equity through: assessing how spatial allocations affect different user groups, prioritizing access for small-scale and traditional users, avoiding concentration of benefits among industrial actors, and establishing benefit-sharing mechanisms for spatial allocations [489].

Participatory MSP processes: Design MSP through inclusive processes that: engage diverse stakeholders from early stages, recognize and integrate local and traditional knowledge, provide capacity support for marginalized groups to participate effectively, and ensure communities have power to shape outcomes, not just provide input [490].

Cumulative impact assessment: Assess cumulative social and ecological impacts of multiple ocean uses rather than evaluating projects in isolation [491]. Consider how spatial allocations interact and compound effects on communities and ecosystems [492].

Spatial justice principles: Apply spatial justice principles ensuring: fair distribution of marine space access across social groups, recognition of diverse spatial claims and meanings, procedural fairness in spatial decision-making, and remediation of historical spatial injustices [493].

9.5 Knowledge Integration and Epistemic Justice

Pluralistic knowledge systems: Recognize and integrate diverse knowledge systems in ocean governance through: respectful protocols for Indigenous knowledge engagement, co-production of knowledge across scientific, traditional, and local systems, funding for community-based monitoring and research, and challenging epistemic hierarchies that privilege Western science [494].

Social science integration: Ensure social sciences are integrated alongside natural sciences in ocean research and policy through: funding for social science research on Blue Economy, social scientists in governance bodies and advisory panels, social impact assessment as standard practice, and interdisciplinary research addressing coupled social-ecological systems [495].

Transparency and accessibility: Make ocean science and governance information accessible through: open data policies with appropriate protections for sensitive information, plain-language communication of research findings, translation into local languages, and capacity building for communities to engage with technical information [496].

9.6 Sectoral Policy Recommendations

Fisheries:

- Prioritize small-scale fisheries in resource allocation and support
- Implement co-management with genuine power-sharing
- Ensure equitable benefit-sharing from fisheries resources
- Protect customary fishing rights and territories
- Address gender inequalities in fisheries value chains

Shipping:

- Eliminate flags of convenience through international cooperation
- Strengthen enforcement of labor and safety standards
- Support seafarer organizing and collective bargaining
- Ensure just transition in maritime decarbonization
- Address mental health and wellbeing of seafarers

Offshore energy:

- Require early and sustained community engagement
- Establish benefit-sharing mechanisms for affected communities
- Ensure just transition for workers from fossil fuel industries
- Integrate offshore energy with fisheries and conservation through participatory MSP
- Assess and mitigate social and cultural impacts

Ports:

- Integrate ports with urban planning and community development
- Ensure decent work standards for port workers
- Address environmental justice concerns in port communities
- Support port worker transitions in automation and decarbonization
- Create participatory governance mechanisms for port development

Tourism:

- Promote community-based and sustainable tourism models
- Ensure equitable distribution of tourism benefits
- Protect coastal communities from displacement and gentrification
- Regulate cruise tourism to maximize local benefits

- Integrate tourism with conservation and cultural preservation

9.7 Financing Just Blue Transitions

Public investment: Prioritize public investment in Blue Economy over private financialization to ensure public interest and accountability [497]. Direct public funds toward: community-based enterprises and cooperatives, small-scale fisheries infrastructure and support, coastal community resilience and adaptation, marine conservation with community benefits, and research and monitoring [498].

Conditionality: Attach strong social and environmental conditionality to public funding and subsidies for Blue Economy activities [499]. Require: decent work standards and labor rights, community consultation and consent, equitable benefit-sharing, environmental sustainability, and independent monitoring and evaluation [500].

Benefit-sharing mechanisms: Establish mechanisms ensuring communities benefit from Blue Economy development through: revenue sharing from resource extraction and use, community development funds from private projects, local employment and procurement requirements, and compensation for impacts and displacement [501].

Subsidy reform: Redirect harmful subsidies (e.g., for industrial fishing fleets) toward sustainable and equitable alternatives (e.g., small-scale fisheries support, ecosystem restoration, community adaptation) [502].

9.8 Monitoring and Evaluation

Social indicators: Develop and monitor social indicators alongside economic and environmental metrics [503]. Include: employment quality (not just quantity), income distribution and poverty in coastal communities, gender equity across maritime sectors, community participation and satisfaction with governance, cultural vitality and maritime heritage, access to marine resources across social groups, and health and wellbeing of ocean-dependent communities [504].

Participatory monitoring: Engage communities in monitoring and evaluation through: community-based monitoring programs, participatory indicator development, citizen science initiatives, and community oversight of compliance and impacts [505].

Independent evaluation: Ensure independent evaluation of Blue Economy policies and projects through: third-party social and environmental impact assessment, academic research and analysis, civil society monitoring and reporting, and transparent public reporting of findings [506].

9.9 International Cooperation and Global Justice

Capacity building: Support capacity building in developing countries for: social science research on Blue Economy, participatory governance mechanisms, community organizing and advocacy, and sustainable ocean management [507].

Technology transfer: Facilitate technology transfer for sustainable ocean use on fair and equitable terms, avoiding intellectual property barriers that perpetuate inequalities [508].

Fair trade and market access: Ensure developing countries have fair access to markets for marine products and benefit equitably from global ocean economy [509].

Climate justice: Recognize historical responsibility of developed countries for ocean degradation and climate impacts [510]. Provide adequate climate finance for adaptation and loss and damage in vulnerable coastal communities [511].

Debt relief: Consider debt relief for developing countries conditional on investments in sustainable and equitable ocean management [512].

9.10 Synthesis: Principles for Just Blue Transitions

Just blue transitions should be guided by principles of:

1. Justice: Prioritizing equity, fairness, and rights across distribution, procedure, and recognition
2. Democracy: Ensuring meaningful participation, accountability, and community empowerment
3. Sustainability: Integrating social, ecological, and economic dimensions
4. Precaution: Acting cautiously in face of uncertainty and potential irreversible harms
5. Solidarity: Building alliances across sectors, regions, and generations
6. Transformation: Pursuing fundamental change in power relations and governance, not just incremental reform

These principles require moving beyond blue growth toward alternative paradigms that center justice, democracy, and sustainability. Blue Sociology provides analytical frameworks and empirical evidence to support this transformation.

---

10. Conclusion: Blue Sociology as Complementary Renewal {#10-conclusion}

10.1 Synthesis of Key Arguments

This working paper has proposed Blue Sociology as a complementary renewal and extension of maritime sociology toward an integrated sociology of sea societies. The argument proceeds through several interconnected claims:

First, contemporary Blue Economy development presents urgent social sustainability challenges—labor precarity, distributional conflicts, participation deficits, power asymmetries, and justice concerns—that existing disciplinary

frameworks inadequately address [513]. Analysis of EU Ocean Pact documents and comprehensive literature synthesis reveals systematic marginalization of social dimensions in Blue Economy research and policy [514].

Second, maritime sociology provides essential foundations through decades of empirical research on maritime labor, organizations, and communities, and analytical traditions in political economy, institutional analysis, and territorial transformation [515]. Janiszewski's maritimization theory and subsequent work on labor regimes, organizational dynamics, and port development offer crucial insights for understanding Blue Economy transformations [516].

Third, the oceanic turn contributes vital theoretical innovations—more-than-human geographies, hydrosocial relations, materialist ontologies, Indigenous epistemologies, and feminist perspectives—that reframe oceans as relational, political, and lived spaces rather than inert resources [517]. These perspectives enable analysis of multispecies entanglements, infrastructural politics, and planetary-scale socioecological dynamics [518].

Fourth, sectoral analysis demonstrates how labor regimes, institutional power, legitimacy dynamics, and justice concerns manifest differently across fisheries, ports, shipping, offshore energy, and tourism, yet common patterns of stratification, precarity, and power asymmetries emerge [519]. Blue Sociology must attend to both sectoral specificities and cross-cutting dynamics [520].

Fifth, systematic research gaps in Blue Economy scholarship—equity marginalization, participation deficits, governance fragmentation, methodological bias toward economic metrics, knowledge exclusion, power asymmetries, and social-ecological integration deficits—can be addressed through maritime social science methods including political ecology, participatory research, multi-level governance analysis, social-ecological systems theory, ethnography, and feminist approaches [521].

Sixth, the tripartite Marine-Maritime-Oceanic analytical framework enables integrated analysis across natural processes, human activities, and planetary systems, examining interconnections and feedbacks that shape contemporary sea societies [522]. Six core analytical dimensions—labor and social reproduction, institutional arrangements and power relations, spatial dynamics and territorial transformations, socioecological relations and multispecies entanglements, knowledge and culture, and justice and democracy—provide organizing architecture for Blue Sociology research [523].

Seventh, policy implications emphasize moving from blue growth to blue justice through governance democratization, labor standards and just transitions, spatial justice in marine spatial planning, knowledge integration and epistemic justice, sectoral reforms, and financing mechanisms that prioritize equity and sustainability [524].

10.2 Blue Sociology's Distinctive Contribution

Blue Sociology's distinctive contribution lies in synthesizing maritime sociology's empirical depth and labor-centered analysis with oceanic turn's theoretical innovations and justice commitments. This synthesis enables analysis that is simultaneously:

- Grounded: Rooted in material conditions of maritime work and coastal livelihoods, drawing on decades of empirical research
- Relational: Attentive to multispecies entanglements, material agencies, and more-than-human dimensions often absent from social science
- Critical: Foregrounding power relations, political economy, and structural inequalities that shape ocean use and governance
- Integrative: Addressing coupled social-ecological systems across sectors, scales, and domains
- Justice-oriented: Explicitly engaging normative questions and supporting struggles for more equitable and democratic ocean futures
- Transformative: Analyzing not just current conditions but pathways toward fundamental change

This combination distinguishes Blue Sociology from: maritime sociology (which it extends but does not replace), ocean governance studies (which often lack social science depth), environmental sociology (which has given limited attention to oceans), and Blue Economy research (which marginalizes social dimensions).

10.3 Complementary Renewal, Not Replacement

It is crucial to emphasize that Blue Sociology is proposed as complementary renewal rather than replacement of maritime sociology. Maritime sociology's empirical traditions, analytical tools, and institutional presence remain valuable and should be preserved and strengthened. Blue Sociology builds on these foundations while:

- Extending sectoral focus to cross-sectoral integration
- Incorporating socioecological and more-than-human perspectives
- Engaging planetary-scale processes and governance
- Deepening justice and democratic governance analysis
- Integrating oceanic turn theoretical innovations

The relationship is analogous to how environmental sociology extended rural sociology and urban sociology without replacing them—adding new dimensions while maintaining continuity with established traditions.

10.4 Challenges and Limitations

Developing Blue Sociology faces several challenges:

Institutional challenges: Maritime sociology has limited institutional presence compared to rural or urban sociology. Building Blue Sociology requires: establishing academic positions and programs, securing research funding, creating

journals and publication venues, and building professional networks and associations.

Interdisciplinary tensions: Integrating across maritime sociology, environmental sociology, human geography, political ecology, and ocean sciences involves navigating different epistemologies, methods, and institutional cultures. Productive interdisciplinarity requires mutual respect and genuine dialogue rather than disciplinary imperialism.

Engagement challenges: Blue Sociology's commitment to participatory research and community engagement requires: long-term relationships and trust-building, resources for sustained engagement, navigating power dynamics in research relationships, and balancing academic and community priorities.

Political challenges: Blue Sociology's critical stance and justice commitments may create tensions with: funding agencies promoting blue growth, government agencies implementing contested policies, and corporate actors whose practices are critiqued. Maintaining critical independence while engaging policy requires careful navigation.

Methodological challenges: Blue Sociology's methodological pluralism requires: researchers with diverse skills and training, resources for intensive qualitative research, integration of quantitative and qualitative approaches, and publication venues that value diverse methods.

10.5 Future Directions

Advancing Blue Sociology requires action on multiple fronts:

Research: Pursuing the research agenda outlined in Section 8, with priority themes including labor regimes and just transitions, governance institutions and democratic legitimacy, spatial justice and territorial transformations, social-ecological dynamics, knowledge politics, blue justice movements, and transformation pathways.

Capacity building: Developing training programs, curricula, and educational resources in Blue Sociology. Supporting early-career researchers and building diverse scholarly communities.

Network building: Creating national and international networks of Blue Sociology researchers. Facilitating knowledge exchange and collaboration across regions and disciplines.

Community engagement: Building partnerships with fishing communities, maritime workers, coastal residents, and social movements. Supporting community-based research and participatory knowledge production.

Policy engagement: Translating research findings for policy audiences. Engaging with governance institutions while maintaining critical independence. Supporting evidence-informed policy development.

Publication and dissemination: Establishing publication venues for Blue Sociology research. Disseminating findings through academic and non-academic channels. Making research accessible to diverse audiences.

Institutional development: Advocating for academic positions, programs, and funding in Blue Sociology. Building institutional infrastructure to support the field's growth.

10.6 Toward Integrated Sociology of Sea Societies

Ultimately, Blue Sociology aims toward an integrated sociology of sea societies—comprehensive understanding of how human societies relate to, depend upon, transform, and are transformed by oceans and seas. This requires:

- Analyzing labor, livelihoods, and social reproduction in maritime contexts
- Examining institutions, power relations, and governance across scales
- Understanding spatial dynamics and territorial transformations
- Addressing socioecological relations and multispecies entanglements
- Engaging knowledge politics and cultural meanings
- Pursuing justice, democracy, and transformation

Such integrated sociology is essential for addressing the profound challenges facing oceans and coastal communities in the 21st century: climate change and ocean acidification, biodiversity loss and ecosystem degradation, overfishing and resource depletion, pollution and plastic accumulation, coastal development and habitat loss, and social inequalities and injustices.

Blue Economy development, if pursued through blue growth paradigms that privilege economic competitiveness over social justice and ecological sustainability, risks exacerbating these challenges. Alternative pathways centered on justice, democracy, and sustainability are possible but require fundamental transformation in how oceans are governed and used.

Blue Sociology provides analytical frameworks, empirical evidence, and methodological tools to support this transformation. By foregrounding labor, power, justice, and socioecological relations, Blue Sociology can contribute to building more equitable, democratic, and sustainable ocean futures.

10.7 Final Reflection

The ocean covers 71% of Earth's surface, regulates climate, produces oxygen, provides food and livelihoods for billions, and holds profound cultural and spiritual significance for coastal and island peoples. Yet oceans have been systematically

marginalized in social science—treated as spaces of transit, sources of resources, or environmental problems rather than as social spaces and lived environments.

Maritime sociology has long challenged this marginalization through sustained attention to seafaring communities, maritime labor, and coastal societies. The oceanic turn has further reframed oceans as relational, political, and meaningful spaces. Blue Sociology builds on these foundations to develop comprehensive frameworks for understanding contemporary sea societies in the context of accelerating Blue Economy development.

This is not merely academic exercise but urgent practical necessity. Decisions made now about ocean governance, Blue Economy development, and marine spatial planning will shape ocean futures for generations. These decisions must be informed by robust social science that foregrounds justice, democracy, and sustainability alongside economic and ecological considerations.

Blue Sociology offers such social science—grounded in empirical research, theoretically sophisticated, methodologically pluralist, justice-oriented, and engaged with struggles for more equitable ocean futures. By synthesizing maritime sociology's strengths with oceanic turn innovations, Blue Sociology provides the analytical architecture for integrated sociology of sea societies that contemporary challenges demand.

The task ahead is substantial: building institutional infrastructure, conducting research, engaging communities and policy, and contributing to transformation toward just and sustainable ocean futures. This working paper provides conceptual foundations and research agenda for this collective endeavor. The work of developing Blue Sociology—as complementary renewal extending maritime sociology toward integrated sociology of sea societies—has begun.

References

[1] Addamo, A. M., et al. (2020). The EU Blue Economy Report 2020. Publications Office of the European Union.

[2] European Commission. (2024). European Ocean Pact. Brussels: European Commission.

[3] Germond-Duret, C., et al. (2022). (In)justice and the blue economy. *The Geographical Journal*, 188(4), 521-535. https://doi.org/10.1111/geoj.12483

[4] Evans, L., et al. (2023). Putting coastal communities at the center of a sustainable blue economy: A review of risks, opportunities, and strategies. *Frontiers in Political Science*, 4, 1032204. https://doi.org/10.3389/fpos.2022.1032204

[5] Sowman, M., et al. (2023). Strategies for addressing conflicts arising from blue growth initiatives: insights from three case studies in South Africa. *Maritime Studies*, 22, 341-1. https://doi.org/10.1007/s40152-023-00341-1

[6] Cohen, P. J., et al. (2019). Securing a Just Space for Small-Scale Fisheries in the Blue Economy. *Frontiers in Marine Science*, 6, 171. https://doi.org/10.3389/FMARS.2019.00171

[7] Ayilu, R. K., et al. (2023). Blue economy: industrialisation and coastal fishing livelihoods in Ghana. *Reviews in Fish Biology and Fisheries*, 33, 749-0. https://doi.org/10.1007/s11160-022-09749-0

[8] Reis-Filho, J. A., et al. (2024). Navigating Blue Justice: Policy Gaps and Conflicts in Coastal Development from Small-Scale Fisher Perspectives. *One Earth*, 7(7). https://doi.org/10.1016/j.oneear.2024.07.014

[9] Gerhardinger, L. C., et al. (2023). Challenging the blue economy: voices from artisanal fishing communities in Latin America and the Caribbean. *Development*, 66, 366-3. https://doi.org/10.1057/s41301-023-00366-3

[10] Kahveci, E., et al. (2006). *The Other Car Workers: Work, Organisation and Technology in the Maritime Car Carrier Industry*. Palgrave Macmillan.

[11] Chin, C. B. N. (2008). Labour Flexibilization at Sea. *International Feminist Journal of Politics*, 10(1), 1-18. https://doi.org/10.1080/14616740701747584

[12] Narchi, N. E., et al. (2018). El CoLaboratorio de oceanografía social: espacio plural para la conservación integral de los mares y las sociedades costeras. *Sociedad y Ambiente*, 18, 1888. https://doi.org/10.31840/SYA.V0I18.1888

[13] Tupara-Katene, T., et al. (2025). Tangaroa Ararau: Reimagining Ocean Governance by Integrating Indigenous Perspectives for a Sustainable Future. *Ocean Sciences*, 1237. https://doi.org/10.5194/oos2025-1237

[14] Baker-Médard, M., et al. (2023). Gender equity and collaborative care in Madagascar's locally managed marine areas: reflections on the launch of a fisherwomen's network. *Ecology and Society*, 28(2), 26. https://doi.org/10.5751/es-13959-280226

[15] Hadjimichael, M. (2018). A call for a blue degrowth: unravelling the European Union's fisheries and maritime policies. *Marine Policy*, 94, 158-164. https://doi.org/10.1016/J.MARPOL.2018.05.007

[16] Bennett, N. J. (2018). Navigating a just and inclusive path towards sustainable oceans. *Marine Policy*, 97, 139-146. https://doi.org/10.1016/J.MARPOL.2018.06.001

[17] Bavinck, M., et al. (2020). Manifesto for the marine social sciences. *Maritime Studies*, 19, 121-123. https://doi.org/10.1007/S40152-020-00179-X

[18] Voyer, M., & van Leeuwen, J. (2019). 'Social license to operate' in the Blue Economy. *Resources Policy*, 62, 102-113. https://doi.org/10.1016/J.RESOURPOL.2019.02.020

[19] Arbo, P., et al. (2018). The transformation of the oceans and the future of marine social science. *Maritime Studies*, 17, 295-304. https://doi.org/10.1007/S40152-018-0117-5

[20] Guerreiro, J. (2021). The Blue Growth Challenge to Maritime Governance. *Frontiers in Marine Science*, 8, 681546. https://doi.org/10.3389/FMARS.2021.681546

[21] Anderson, J., & Peters, K. (2016). *Water Worlds: Human Geographies of the Ocean*. Routledge.

[22] Peters, K. (2020). The territories of governance: unpacking the ontologies and geophilosophies of fixed to flexible ocean management, and beyond. *Philosophical Transactions of the Royal Society B*, 375, 20190458. https://doi.org/10.1098/RSTB.2019.0458

[23] Bear, C. (2017). Assembling ocean life: more-than-human entanglements in the Blue economy. *Dialogues in Human Geography*, 7(1), 27-31. https://doi.org/10.1177/2043820617691635

[24] Hannigan, J. (2017). Toward a Sociology of Oceans. *Canadian Review of Sociology*, 54(1), 8-27. https://doi.org/10.1111/CARS.12136

[25] Longo, S. B., et al. (2016). An Ocean of Troubles: Advancing Marine Sociology. *Social Problems*, 63(4), 463-479. https://doi.org/10.1093/SOCPRO/SPW023

[26] Blythe, J., et al. (2023). Blue justice: a review of emerging scholarship and resistance movements. *Cambridge Prisms: Coastal Futures*, 1, e4. https://doi.org/10.1017/cft.2023.4

[27] Ertor, I. (2021). 'We Are the Oceans, We Are the People!': Fisher People's Struggles for Blue Justice. *Journal of Peasant Studies*, 48(7), 1-32. https://doi.org/10.1080/03066150.2021.1999932

[28] Clark, B. (2022). Racial capitalism and the sea: Development and change in Black maritime labour, and what it means for fisheries and a blue economy. *Fish and Fisheries*, 23(3), 639-653. https://doi.org/10.1111/faf.12639

[29] Bavinck, M., et al. (2018). Fisheries as social struggle: a reinvigorated social science research agenda. *Marine Policy*, 94, 46-52. https://doi.org/10.1016/J.MARPOL.2018.04.026

[30] Campling, L., & Colás, A. (2018). Capitalism and the sea: Sovereignty, territory and appropriation in the global ocean. *Environment and Planning D: Society & Space*, 36(4), 776-794. https://doi.org/10.1177/0263775817737319

[31] Poole, M. (1981). Maritime sociology: towards a delimation of themes and analytical frameworks. *Maritime Policy & Management*, 8(1), 1-14. https://doi.org/10.1080/03088838100000046

[32] Sowa, F., et al. (2015). A Short Introductory Note on Maritime Sociology. *International Journal of Maritime History*, 27(4), 813-814.

[33] Cocco, G. (2013). Theoretical Implications of Maritime Sociology. *International Journal of Maritime History*, 25(2), 293-306.

[34] Lamvik, G. M. (2012). The Filipino seafarer: a life between sacrifice and shopping. *Anthropology in Action*, 19(1), 22-38. https://doi.org/10.3167/AIA.2012.190104

[35] Kołodziej-Durnaś, A. (2014). Współcześni polscy marynarze – kręgi izolacji. *Przegląd Socjologii Jakościowej*, 10(3), 6-27.

[36] Fricke, P. H. (1974). *The social structure of crews of British dry cargo merchant ships: A study of the organization and environment of an occupation*. PhD thesis, University of Liverpool.

[37] Clark, M. A. (2016). *Merchant Marine Deck Officer Agency Through Performative Acts*. PhD thesis, Cardiff University.

[38] Broeze, F. (2007). Maritime Labour. In *Maritime labour: contributions to the history of work at sea, 1500-2000* (pp. 153-174). Liverpool University Press.

[39] Boewe, J. (2023). Major trends in work at sea: outline of a political economy of maritime labour. In *Research Handbook on the Sociology of Organizations* (pp. 63-78). Edward Elgar Publishing. https://doi.org/10.4337/9781839106583.00063

[40] Alderton, T., et al. (2004). *The global seafarer: living and working conditions in a globalized industry*. International Labour Organization.

[41] Ruggunan, S. (2008). *Global transformation of the contemporary labour market for merchant navy seafarers: case studies of Filipino, South African and British seafaring labour markets*. PhD thesis, University of KwaZulu-Natal.

[42] Ruggunan, S. (2011). The Role of Organised Labour in Preventing a 'Race to the Bottom' for Filipino Seafarers in the Global Labour Market. *African and Asian Studies*, 10(2-3), 133-159. https://doi.org/10.1163/156921011X587022

[43] Morris, R., & Devlin, P. (2001). Crews of convenience from the South West Pacific: the 'German' sailors of Kiribati. *New Zealand Journal of Industrial Relations*, 26(3), 259-273.

[44] Liang, Y. (2011). *Sailing on a Neoliberal Sea: Multinational Seafarers on Container Ships*. PhD thesis, Cardiff University.

[45] Vigarie, A. (1997). L'évolution des sociétés portuaires et la démaritimisation des villes ports en Europe continentale du nord-ouest. *Norois*, 176, 627-641.

[46] Ibid.

[47] Giolito, V., et al. (2021). La maritimización y su huella territorial: una trayectoria sociotécnica.: El caso de la Bahía de Pasaia (País Vasco, España). *Finisterra*, 56(118), 23-46. https://doi.org/10.18055/FINIS23746

[48] Foulquier, E. (2025). La maritimité qui vient, une occupation industrielle du monde océanique. *Géocarrefour*, 99(1). https://doi.org/10.4000/1448d

[49] Pistone, F. (2024). The Contemporary Coast as an Urban Amphibious. The Complex Relationship Between City–Sea Interface and Urban Coastal Society After the COVID-19 Crisis. In *The Urban Book Series* (pp. 67-82). Springer. https://doi.org/10.1007/978-3-031-56607-3_5

[50] Nenadić, M., et al. (2016). Sociological aspects of seafarers' life and work and management styles in shipping. *Naše More*, 63(3), 119-125.

[51] Walters, D., & Bailey, N. (2013). *Lives in Peril: Profit or Safety in the Global Maritime Industry?* Palgrave Macmillan.

[52] Ibid.

[53] Kerr, S., et al. (2018). Shaping Blue Growth: Social Sciences at the Nexus Between Marine Renewables and Energy Policy. In *The Palgrave Handbook of Managing Fossil Fuels and Energy Transitions* (pp. 41-66). Springer. https://doi.org/10.1007/978-3-319-99097-2_3

[54] Krátká, L. (2020). Life like a swing: women's perspectives of everyday life in czechoslovak seafarers' families under state socialism. *International Journal of Maritime History*, 32(1), 246-262. https://doi.org/10.26774/WRHM.246

[55] Kołodziej, E., & Kołodziej-Durnaś, A. (2012). Determinanty więzi społecznej na statku – badania jakościowe wśród marynarzy. *Przegląd Socjologiczny*, 61(2), 155-178.

[56] Ibid.

[57] Harris, J. L., & Thompson, B. S. (2023). Supporting places left to the sea: A place-based research agenda for regional coastal transformations in the blue economy. *Progress in Environmental Geography*, 2(3-4), 212672. https://doi.org/10.1177/27539687231212672

[58] Janiszewski, L. (1934). *Maritimizacja*. Gdynia: Instytut Bałtycki.

[59] Giolito et al. (2021), op. cit.

[60] Foulquier (2025), op. cit.

[61] Ibid.

[62] Campling & Colás (2018), op. cit.

[63] Boewe (2023), op. cit.

[64] Alderton et al. (2004), op. cit.

[65] Walters & Bailey (2013), op. cit.

[66] Chin (2008), op. cit.

[67] Ruggunan (2011), op. cit.

[68] Morris & Devlin (2001), op. cit.

[69] Lamvik (2012), op. cit.

[70] Clark (2022), op. cit.

[71] Walters & Bailey (2013), op. cit.

[72] Ibid.

[73] Boewe (2023), op. cit.

[74] Gekara, V. (2021). Union Organising in the Context of Regional Labour Market Decline: The Case of Nautilus International. In *Trade Unions and Migrant Workers* (pp. 267-286). Springer. https://doi.org/10.1007/978-3-030-49825-2_13

[75] Bailey, N., et al. (2018). Framing social justice: the ties that bind a multinational occupational community. *Sociology*, 52(2), 348-364. https://doi.org/10.1177/0038038516670111

[76] Gekara (2021), op. cit.

[77] Boewe (2023), op. cit.

[78] Longo et al. (2016), op. cit.

[79] Arbo et al. (2018), op. cit.

[80] Longo et al. (2016), op. cit.

[81] Arbo et al. (2018), op. cit.

[82] Martín-González, J. M. (2021). The 'Oceanic Turn' and the Indian Ocean: Maritime Criticism, Cosmopolitanism and Nineteenth-Century Oceanic Culture. In *The Palgrave Handbook of Global Oceanic Literatures* (pp. 17-34). Springer. https://doi.org/10.1007/978-3-030-77056-3_2

[83] Peters, K., et al. (2019). Oceanic Travels: Future Voyages for Moving Deep and Wide within the "New Mobilities Paradigm". *Transfers*, 9(2), 70-91. https://doi.org/10.3167/TRANS.2019.090207

[84] Anderson & Peters (2016), op. cit.

[85] Peters, K., et al. (2022). *The Routledge handbook of ocean space*. Routledge. https://doi.org/10.4324/9781315111643

[86] Hannigan (2017), op. cit.

[87] Bear (2017), op. cit.

[88] Crowder, L. B., et al. (2019). The changing social world of the oceans. In *Predicting Future Oceans* (pp. 293-302). Elsevier. https://doi.org/10.1016/B978-0-12-817945-1.00025-3

[89] Bear (2017), op. cit.

[90] Ibid.

[91] Ibid.

[92] Martin, K. S., et al. (2017). Creating Space for Community in Marine Conservation and Management: Mapping "Communities-at-Sea". In *Conservation for the Anthropocene Ocean* (pp. 123-141). Elsevier. https://doi.org/10.1016/B978-0-12-805375-1.00007-6

[93] Ibid.

[94] Bear (2017), op. cit.

[95] Kerr et al. (2018), op. cit.

[96] Ibid.

[97] Ibid.

[98] Bear (2017), op. cit.

[99] Ibid.

[100] Peters (2020), op. cit.

[101] Bruyns, G., & Graafland, A. (2017). Aquatic Urbanisms: Water as Planning and Territorial Instrument Considering The 9 Dash Line Policy. *IN_BO*, 8(11), 42-53. https://doi.org/10.6092/2531-9906/5420

[102] Peters (2020), op. cit.

[103] Ibid.

[104] Ibid.

[105] Bruyns & Graafland (2017), op. cit.

[106] Peters (2020), op. cit.

[107] Ibid.

[108] Ibid.

[109] Makoni, S., et al. (2023). Waterscape Epistemologies, Waves of Knowing and Translanguaging as Wet Ontology. *Applied Linguistics*, 44(5), 789-808.

[110] Ibid.

[111] Peters (2020), op. cit.

[112] Ibid.

[113] Ibid.

[114] Ibid.

[115] Kahveci et al. (2006), op. cit.

[116] Mevik, K. (2024). Maritime Skills and the (Not So) Changing Qualification Landscape in Global Shipping. *Public Anthropologist*, 6(1), 65-87. https://doi.org/10.1163/25891715-bja10065

[117] Ibid.

[118] Peters (2020), op. cit.

[119] Ibid.

[120] Levain, A., et al. (2020). Green Out of the Blue, or How (Not) to Deal with Overfed Oceans: An Analytical Review of Coastal Eutrophication and Social Conflict. *Anthropological Review*, 11(1), 108-130. https://doi.org/10.3167/ARES.2020.110108

[121] Ibid.

[122] Beatley, T. (2014). *Blue Urbanism: Exploring Connections Between Cities & Oceans*. Island Press.

[123] Anbleyth-Evans, J., et al. (2023). Urban Marine Protected Areas and the Role of Blue Democracy Towards Healthy Coastal Cities. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4516616

[124] Pistone (2024), op. cit.

[125] Ibid.

[126] Pittman, J., et al. (2019). Marine parks for coastal cities: A concept for enhanced community well-being, prosperity and sustainable city living. *Marine Policy*, 103, 160-171. https://doi.org/10.1016/J.MARPOL.2019.02.012

[127] Anbleyth-Evans et al. (2023), op. cit.

[128] Ibid.

[129] Pistone (2024), op. cit.

[130] Ibid.

[131] Ibid.

[132] Ibid.

[133] Dahou, T., & Ould Cheikh, A. W. (2021). La mer, un construit politique. La privatisation des territoires et des ressources maritimes en acte. *VertigO*, 21(1). https://doi.org/10.4000/VERTIGO.29593

[134] Ibid.

[135] Mallin, F. (2025). Blue Economy Struggles—Capital and Power in the Global Ocean: Introduction. *Journal of Agrarian Change*, 25(1), e70014. https://doi.org/10.1111/joac.70014

[136] Ibid.

[137] Dahou & Ould Cheikh (2021), op. cit.

[138] Ibid.

[139] Hou, Y. (2023). Social Construction of the Ocean, Sea Power, and Maritime Transformation. *Journal of Political & Military Sociology*, 50(1), 1-20. https://doi.org/10.5744/jpms.2023.1002

[140] Ibid.

[141] Peters (2020), op. cit.

[142] Ibid.

[143] Tupara-Katene et al. (2025), op. cit.

[144] Hunter, L., et al. (2024). Climate change and oceanic responsibilities: Listening and dancing with Saltwater Country, Australia. In *Indigenous Peoples, Climate Change and the Law* (pp. 145-162). Bristol University Press. https://doi.org/10.56687/9781529235319-009

[145] Tupara-Katene et al. (2025), op. cit.

[146] Ibid.

[147] Hunter et al. (2024), op. cit.

[148] Ibid.

[149] Tupara-Katene et al. (2025), op. cit.

[150] Ibid.

[151] Ibid.

[152] Ibid.

[153] Ibid.

[154] Baker-Médard et al. (2023), op. cit.

[155] Chin (2008), op. cit.

[156] Baker-Médard et al. (2023), op. cit.

[157] Ibid.

[158] Ibid.

[159] Ibid.

[160] Chin (2008), op. cit.

[161] Ibid.

[162] Baker-Médard et al. (2023), op. cit.

[163] Ibid.

[164] European Commission (2024), op. cit.

[165] Ibid.

[166] Ibid.

[167] Ibid.

[168] Ibid.

[169] Ibid.

[170] Ibid.

[171] Ibid.

[172] Ibid.

[173] Ibid.

[174] Ibid.

[175] Analysis based on EU Ocean Pact document review.

[176] European Commission (2024), op. cit.

[177] Analysis based on EU Ocean Pact document review.

[178] European Commission (2024), op. cit.

[179] Ibid.

[180] Analysis based on EU Ocean Pact document review.

[181] European Commission (2024), op. cit.

[182] Ibid.

[183] Analysis based on EU Ocean Pact document review.

[184] Ibid.

[185] Ibid.

[186] Ibid.

[187] Ibid.

[188] European Commission (2024), op. cit.

[189] Analysis based on EU Ocean Pact document review.

[190] Ibid.

[191] Ibid.

[192] Ibid.

[193] Ibid.

[194] Ibid.

[195] European Commission (2024), op. cit.

[196] Analysis based on EU Ocean Pact document review.

[197] European Commission (2024), op. cit.

[198] Analysis based on EU Ocean Pact document review.

[199] Ibid.

[200] European Commission (2024), op. cit.

[201] Analysis based on EU Ocean Pact document review.

[202] Ibid.

[203] Ibid.

[204] European Commission (2024), op. cit.

[205] Analysis based on EU Ocean Pact document review.

[206] Ibid.

[207] Ibid.

[208] European Commission (2024), op. cit.

[209] Analysis based on EU Ocean Pact document review.

[210] Ibid.

[211] Ibid.

[212] Ibid.

[213] Ibid.

[214] Ibid.

[215] Ibid.

[216] Ibid.

[217] Ibid.

[218] Ibid.

[219] Ibid.

[220] Ibid.

[221] Ibid.

[222] Ibid.

[223] European Commission (2024), op. cit.

[224] Analysis based on EU Ocean Pact document review.

[225] Ibid.

[226] Ibid.

[227] Ibid.

[228] Ibid.

[229] Ibid.

[230] Ibid.

[231] European Commission (2024), op. cit.

[232] Analysis based on EU Ocean Pact document review.

[233] Ibid.

[234] Ibid.

[235] Ibid.

[236] Ibid.

[237] Ibid.

[238] Ibid.

[239] Ibid.

[240] Ibid.

[241] Ibid.

[242] Ibid.

[243] Ibid.

[244] Ibid.

[245] Ibid.

[246] Ibid.

[247] Ibid.

[248] Ibid.

[249] Addamo et al. (2020), op. cit.

[250] Synthesis based on comprehensive literature review.

[251] Cohen et al. (2019), op. cit.

[252] Ayilu et al. (2023), op. cit.

[253] Clark (2022), op. cit.

[254] Cohen et al. (2019), op. cit.

[255] Ayilu et al. (2023), op. cit.

[256] Caswell, B. A., et al. (2020). Something old, something new: historical perspectives provide lessons for blue growth agendas. *Fish and Fisheries*, 21(4), 774-796. https://doi.org/10.1111/FAF.12460

[257] Ibid.

[258] Carothers, C. (2011). Equity and access to fishing rights: exploring the community quota program in the Gulf of Alaska. *Human Organization*, 70(3), 255-264. https://doi.org/10.17730/HUMO.70.3.D686U2R7J2267055

[259] Ibid.

[260] Walters & Bailey (2013), op. cit.

[261] Clark et al. (2019). From Sea Slaves to Slime Lines: Commodification and Unequal Ecological Exchange in Global Marine Fisheries. In *Ecologically Unequal Exchange* (pp. 163-188). Springer. https://doi.org/10.1007/978-3-319-89740-0_8

[262] Cohen et al. (2019), op. cit.

[263] Baker-Médard et al. (2023), op. cit.

[264] Ibid.

[265] Kleiber, D., et al. (2017). Promoting gender equity and equality through the small-scale fisheries guidelines: experiences from multiple case studies. In *The Small-Scale Fisheries Guidelines* (pp. 737-759). Springer. https://doi.org/10.1007/978-3-319-55074-9_35

[266] Ayilu et al. (2023), op. cit.

[267] Campos, L., et al. (2024). Reconversion or Exclusion? The Effects of Blue Economy Policies on Semi-industrial and Artisanal Fishing in Puntarenas, Costa Rica. *Latin American Perspectives*, 51(6), 1285386. https://doi.org/10.1177/0094582x241285386

[268] Kidd, S., et al. (2019). Taking account of Land-Sea Interactions in Marine Spatial Planning. In *Maritime Spatial Planning* (pp. 245-270). Springer. https://doi.org/10.1007/978-3-319-98696-8_11

[269] Pistone (2024), op. cit.

[270] Ayilu et al. (2023), op. cit.

[271] Blythe et al. (2023), op. cit.

[272] Ertor (2021), op. cit.

[273] Cohen et al. (2019), op. cit.

[274] Gerhardinger et al. (2023), op. cit.

[275] Cohen et al. (2019), op. cit.

[276] Gerhardinger et al. (2023), op. cit.

[277] Ertor (2021), op. cit.

[278] Blythe et al. (2023), op. cit.

[279] Giolito et al. (2021), op. cit.

[280] Vigarie (1997), op. cit.

[281] Kahveci et al. (2006), op. cit.

[282] Vigarie (1997), op. cit.

[283] Foulquier, E. (2012). ¿un espacio excepcional o un bien común? Discutiendo la deficiencia en el gobierno del puerto. *Scripta Nova*, 16(418).

[284] Broeze (2007), op. cit.

[285] Ibid.

[286] Ibid.

[287] Ibid.

[288] Ibid.

[289] Gekara (2021), op. cit.

[290] Foulquier (2012), op. cit.

[291] Ibid.

[292] Ibid.

[293] Ibid.

[294] Ibid.

[295] Vaca-Cabrero, M. (2025). Development of sustainable ports through the Blue Economy and the use of emerging technology tool. PhD thesis, Universidad Politécnica de Madrid. https://doi.org/10.20868/upm.thesis.89961

[296] Ibid.

[297] Ibid.

[298] Alderton et al. (2004), op. cit.

[299] Ruggunan (2008), op. cit.

[300] Lamvik (2012), op. cit.

[301] Ruggunan (2011), op. cit.

[302] Ibid.

[303] Lamvik (2012), op. cit.

[304] Clark (2022), op. cit.

[305] Lamvik (2012), op. cit.

[306] Kołodziej-Durnaś (2014), op. cit.

[307] Chin (2008), op. cit.

[308] Walters & Bailey (2013), op. cit.

[309] Devereux, L., et al. (2022). Forgotten keyworkers: the experiences of British seafarers during the COVID-19 pandemic. *Economic and Labour Relations Review*, 33(1), 79-136. https://doi.org/10.1177/10353046221079136

[310] Ibid.

[311] Walters & Bailey (2013), op. cit.

[312] Nenadić et al. (2016), op. cit.

[313] Walters & Bailey (2013), op. cit.

[314] Nenadić et al. (2016), op. cit.

[315] Walters & Bailey (2013), op. cit.

[316] Boewe (2023), op. cit.

[317] Gekara (2021), op. cit.

[318] Bailey et al. (2018), op. cit.

[319] Gekara (2021), op. cit.

[320] Addamo et al. (2020), op. cit.

[321] Vaca-Cabrero (2025), op. cit.

[322] Ibid.

[323] Kerr et al. (2018), op. cit.

[324] Ibid.

[325] Ibid.

[326] Ibid.

[327] Ibid.

[328] Ibid.

[329] Kidd et al. (2019), op. cit.

[330] Stalmokaitė, I., et al. (2023). Exploring social justice in marine spatial planning: planner and stakeholder perspectives and experiences in the Baltic Sea Region. *Journal of Environmental Planning and Management*, 66(14), 2795-2817. https://doi.org/10.1080/09640568.2023.2279512

[331] Kerr et al. (2018), op. cit.

[332] Ibid.

[333] Ibid.

[334] Ibid.

[335] Ibid.

[336] Ibid.

[337] Mereu, S., et al. (2025). Public Perception and Acceptance of Floating Offshore Wind Energy in Sardinia. In *2025 IEEE International Conference on Clean Electrical Power (ICCEP)* (pp. 1-6). IEEE. https://doi.org/10.1109/iccep65222.2025.11143627

[338] Kerr et al. (2018), op. cit.

[339] Ibid.

[340] Ibid.

[341] Ibid.

[342] Ibid.

[343] Ibid.

[344] Ibid.

[345] Addamo et al. (2020), op. cit.

[346] Chen, C. L., et al. (2020). Blue Economy: Community Case Studies Addressing the Poverty–Environment Nexus in Ocean and Coastal Management. *Sustainability*, 12(11), 4654. https://doi.org/10.3390/SU12114654

[347] Ibid.

[348] Ibid.

[349] Pistone (2024), op. cit.

[350] Alempijević, N., et al. (2019). Nautical Tourism and Small Shipbuilding as Significant part of Blue Economy Development. *Pomorski Zbornik*, 57(1), 107-124. https://doi.org/10.18048/2019.57.07

[351] Ibid.

[352] Ibid.

[353] Chen et al. (2020), op. cit.

[354] Ibid.

[355] Baker-Médard et al. (2023), op. cit.

[356] Chen et al. (2020), op. cit.

[357] Ibid.

[358] Ibid.

[359] Ibid.

[360] Ibid.

[361] Synthesis based on comprehensive literature review.

[362] Ibid.

[363] Ibid.

[364] Ibid.

[365] Ibid.

[366] Ibid.

[367] Ibid.

[368] Ibid.

[369] Ibid.

[370] Kidd et al. (2019), op. cit.

[371] Stalmokaitė et al. (2023), op. cit.

[372] Cohen et al. (2019), op. cit.

[373] Giolito et al. (2021), op. cit.

[374] Alderton et al. (2004), op. cit.

[375] Kerr et al. (2018), op. cit.

[376] Chen et al. (2020), op. cit.

[377] Synthesis based on blue economy gaps analysis.

[378] Ibid.

[379] Evans et al. (2023), op. cit.

[380] Cohen et al. (2019), op. cit.

[381] Ibid.

[382] Ibid.

[383] Ibid.

[384] Ayilu et al. (2023), op. cit.

[385] Cohen et al. (2019), op. cit.

[386] Ayilu et al. (2023), op. cit.

[387] Cohen et al. (2019), op. cit.

[388] Ayilu et al. (2023), op. cit.

[389] Ibid.

[390] Blythe et al. (2023), op. cit.

[391] Sowman et al. (2023), op. cit.

[392] Bennett (2018), op. cit.

[393] Ibid.

[394] Sowman et al. (2023), op. cit.

[395] Bennett (2018), op. cit.

[396] Sowman et al. (2023), op. cit.

[397] Ibid.

[398] Bavinck et al. (2020), op. cit.

[399] Stalmokaitė et al. (2023), op. cit.

[400] Cohen et al. (2019), op. cit.

[401] Sowman et al. (2023), op. cit.

[402] Ibid.

[403] Fidelman, P., et al. (2012). Governing large-scale marine commons: contextual challenges in the Coral Triangle. *Marine Policy*, 36(1), 42-53. https://doi.org/10.1016/J.MARPOL.2011.03.007

[404] Ibid.

[405] Dahou & Ould Cheikh (2021), op. cit.

[406] Fidelman et al. (2012), op. cit.

[407] Harris & Thompson (2023), op. cit.

[408] Fidelman et al. (2012), op. cit.

[409] Ibid.

[410] Harris & Thompson (2023), op. cit.

[411] Fidelman et al. (2012), op. cit.

[412] Ibid.

[413] Mohammed, E. Y., et al. (2017). Fiscal reforms for sustainable marine fisheries governance: Delivering the SDGs and ensuring no one is left behind. *Marine Policy*, 93, 262-270. https://doi.org/10.1016/J.MARPOL.2017.05.017

[414] Evans et al. (2023), op. cit.

[415] Bennett (2018), op. cit.

[416] Ibid.

[417] Evans et al. (2023), op. cit.

[418] Bennett (2018), op. cit.

[419] Evans et al. (2023), op. cit.

[420] Bennett (2018), op. cit.

[421] Ibid.

[422] Evans et al. (2023), op. cit.

[423] Ibid.

[424] Ibid.

[425] Gerhardinger et al. (2023), op. cit.

[426] Ayilu et al. (2023), op. cit.

[427] Ibid.

[428] Gerhardinger et al. (2023), op. cit.

[429] Ayilu et al. (2023), op. cit.

[430] Gerhardinger et al. (2023), op. cit.

[431] Ayilu et al. (2023), op. cit.

[432] Ibid.

[433] Tupara-Katene et al. (2025), op. cit.

[434] Hunter et al. (2024), op. cit.

[435] Tupara-Katene et al. (2025), op. cit.

[436] Baker-Médard et al. (2023), op. cit.

[437] Voyer & van Leeuwen (2019), op. cit.

[438] Sowman et al. (2023), op. cit.

[439] Ibid.

[440] Voyer & van Leeuwen (2019), op. cit.

[441] Sowman et al. (2023), op. cit.

[442] Voyer & van Leeuwen (2019), op. cit.

[443] Sowman et al. (2023), op. cit.

[444] Ibid.

[445] Voyer & van Leeuwen (2019), op. cit.

[446] Ibid.

[447] Sowman et al. (2023), op. cit.

[448] Arbo et al. (2018), op. cit.

[449] Ibid.

[450] Ibid.

[451] Longo et al. (2016), op. cit.

[452] Arbo et al. (2018), op. cit.

[453] Ibid.

[454] Ibid.

[455] Ibid.

[456] Cohen et al. (2019), op. cit.

[457] Ibid.

[458] Ibid.

[459] Arbo et al. (2018), op. cit.

[460] Bavinck et al. (2020), op. cit.

[461] Lamvik (2012), op. cit.

[462] Ibid.

[463] Campling & Colás (2018), op. cit.

[464] Ibid.

[465] Fidelman et al. (2012), op. cit.

[466] Ibid.

[467] Sowman et al. (2023), op. cit.

[468] Ibid.

[469] Arbo et al. (2018), op. cit.

[470] Ibid.

[471] Stalmokaitė et al. (2023), op. cit.

[472] Ibid.

[473] Caswell et al. (2020), op. cit.

[474] Ibid.

[475] Evans et al. (2023), op. cit.

[476] Ibid.

[477] Germond-Duret et al. (2022), op. cit.

[478] Blythe et al. (2023), op. cit.

[479] Fidelman et al. (2012), op. cit.

[480] Ibid.

[481] Bennett (2018), op. cit.

[482] Ibid.

[483] Ibid.

[484] Blythe et al. (2023), op. cit.

[485] Walters & Bailey (2013), op. cit.

[486] Kerr et al. (2018), op. cit.

[487] Baker-Médard et al. (2023), op. cit.

[488] Chin (2008), op. cit.

[489] Stalmokaitė et al. (2023), op. cit.

[490] Ibid.

[491] Kidd et al. (2019), op. cit.

[492] Ibid.

[493] Stalmokaitė et al. (2023), op. cit.

[494] Tupara-Katene et al. (2025), op. cit.

[495] Bavinck et al. (2020), op. cit.

[496] Ibid.

[497] Hadjimichael (2018), op. cit.

[498] Ibid.

[499] Mohammed et al. (2017), op. cit.

[500] Ibid.

[501] Kerr et al. (2018), op. cit.

[502] Mohammed et al. (2017), op. cit.

[503] Evans et al. (2023), op. cit.

[504] Ibid.

[505] Bavinck et al. (2020), op. cit.

[506] Ibid.

[507] Fidelman et al. (2012), op. cit.

[508] Ibid.

[509] Clark (2022), op. cit.

[510] Hunter et al. (2024), op. cit.

[511] Ibid.

[512] Mohammed et al. (2017), op. cit.

[513] Synthesis of comprehensive literature review.

[514] Analysis based on EU Ocean Pact document review.

[515] Poole (1981), op. cit.

[516] Giolito et al. (2021), op. cit.

[517] Anderson & Peters (2016), op. cit.

[518] Peters (2020), op. cit.

[519] Synthesis based on comprehensive literature review.

[520] Ibid.

[521] Synthesis based on blue economy gaps analysis.

[522] Proposed analytical framework.

[523] Ibid.

[524] Policy implications synthesis.

Word Count: ~45,000 words

Date: February 12, 2026

Status: Working Paper for Manuscript Development

This research synthesis document integrates findings from comprehensive literature review across 763 unique papers and policy analysis of EU Ocean Pact documents. It is designed to support development of academic manuscripts, policy briefs, and research proposals advancing Blue Sociology as a complementary renewal of maritime sociology toward integrated sociology of sea societies.